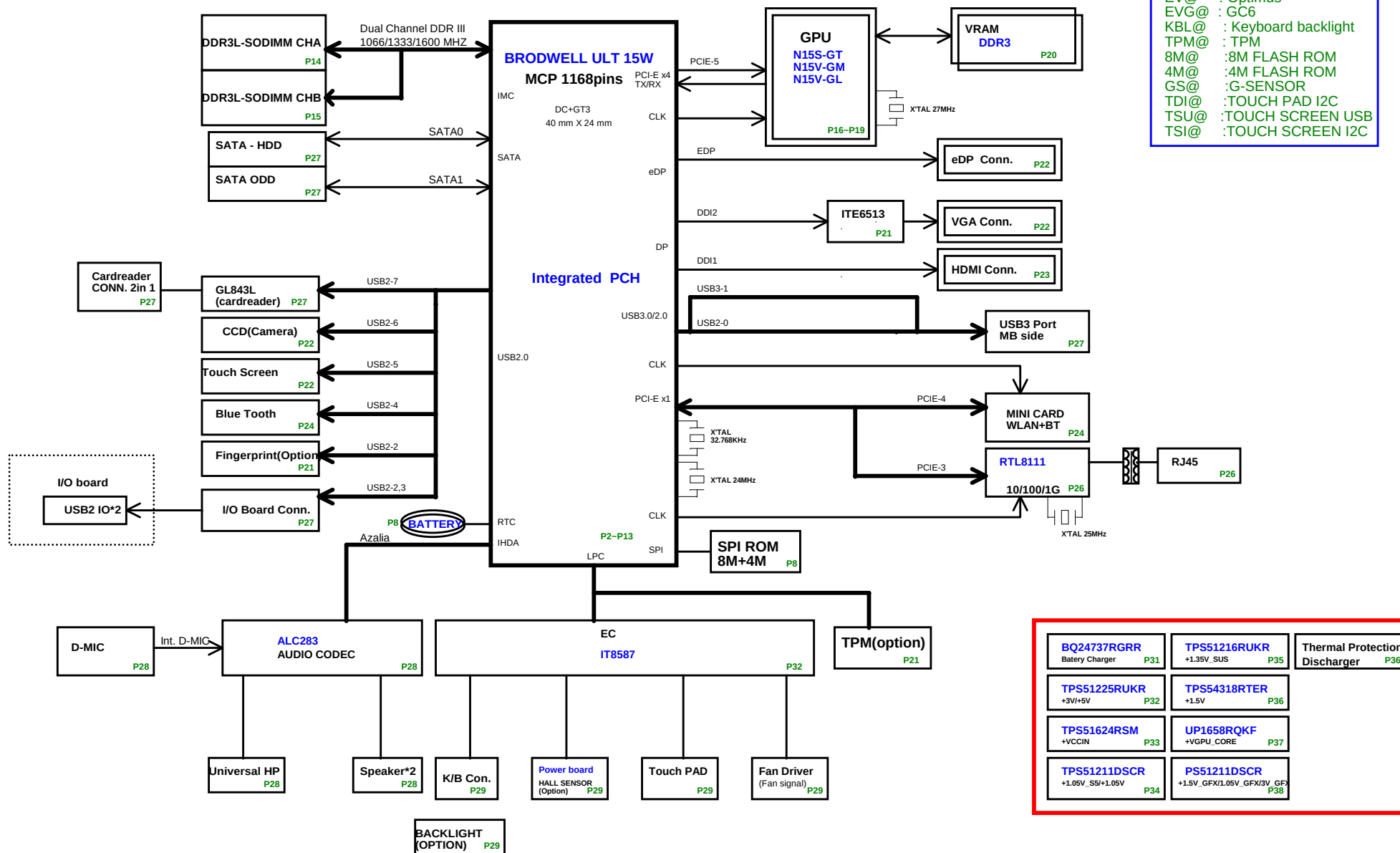


ZQ0 GDDR3 BWD ULT SYSTEM BLOCK DIAGRAM

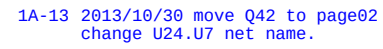
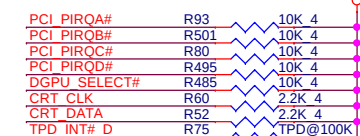
BOM

01

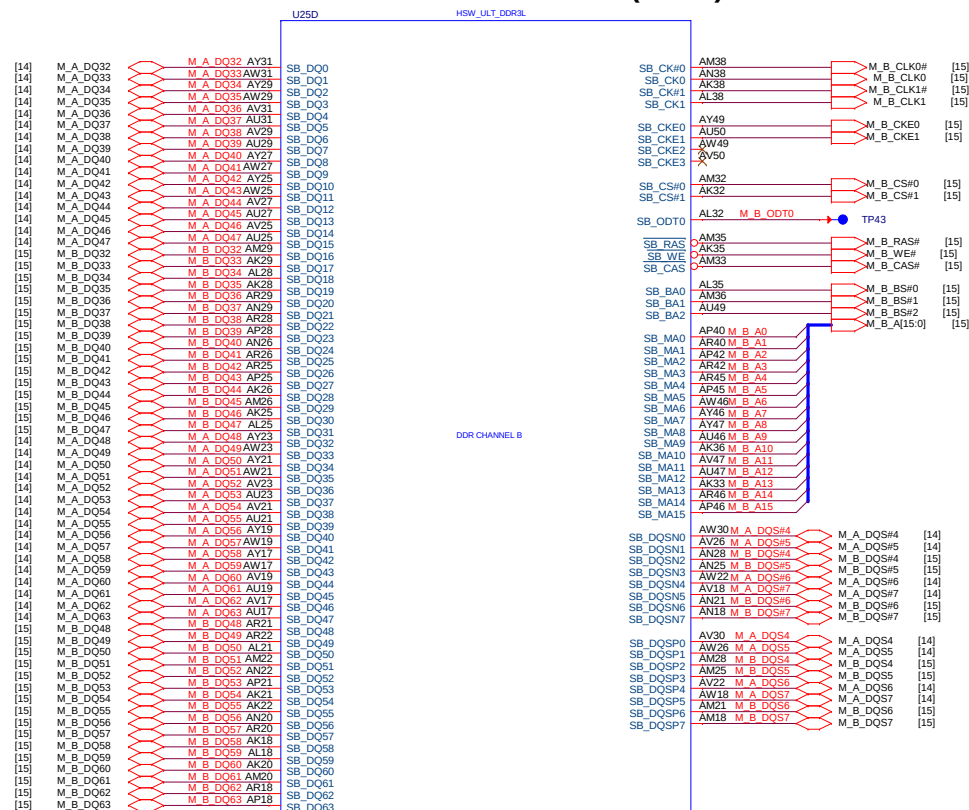


BQ24737RGRR Battery Charger P31	TPS51216RUKR +1.35V_SUS P35	Thermal Protection Discharger P36
TPS51225RUKR +3V/+5V P32	TPS54318RTER +1.5V P36	
TPS51624RSM +VCCIN P33	UP1658RQKF +VGPU_CORE P37	
TPS51211DSCR +1.05V_SS/+1.05V P34	PS51211DSCR +1.5V_GFX/1.05V_GFX/3V_GF P38	

02



Haswell Processor (DDR3)

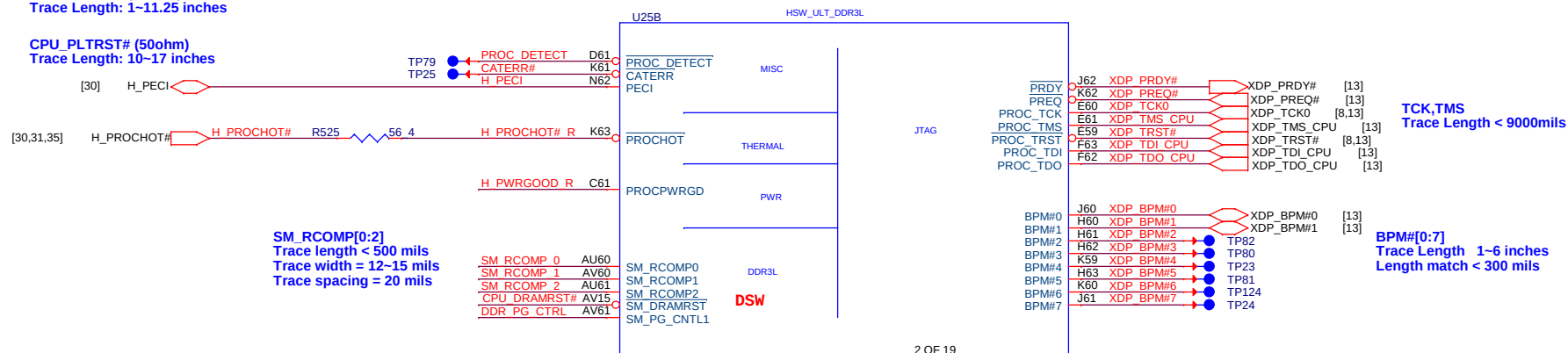


Haswell ULT (SIDEBAND)

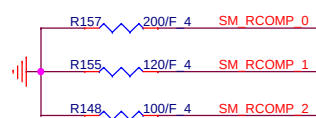
H_PECI (50ohm)
Route on microstrip only
Spacing >18 mils
Trace Length: 0.4~6.125 inches

H_PWRGOOD (50ohm)
Trace Length: 1~11.25 inches

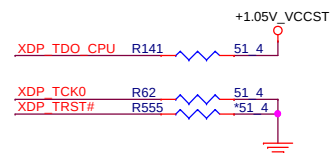
CPU_PLTRST# (50ohm)
Trace Length: 10~17 inches



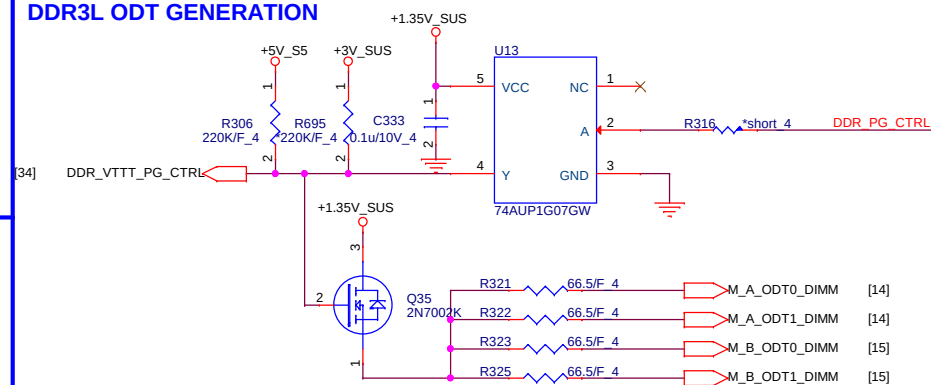
DRAM COMP



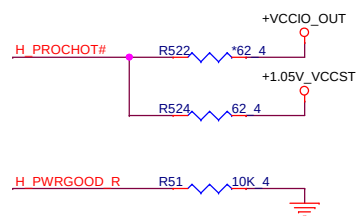
XDP PU/PD



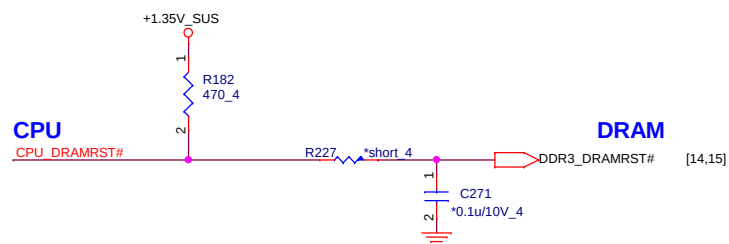
DDR3L ODT GENERATION



PU/PD of CPU



DRAMRST

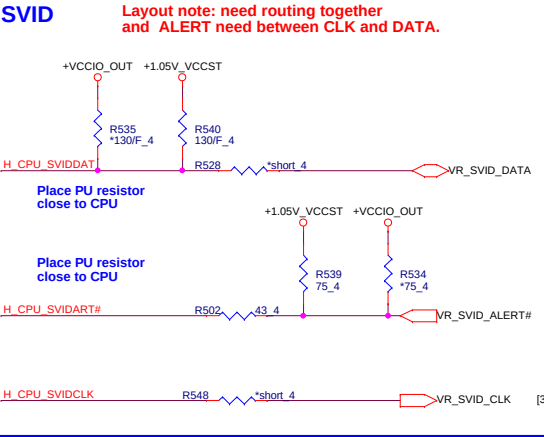
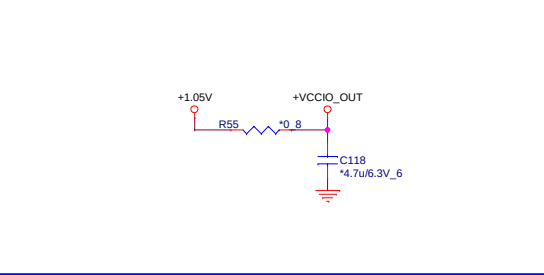
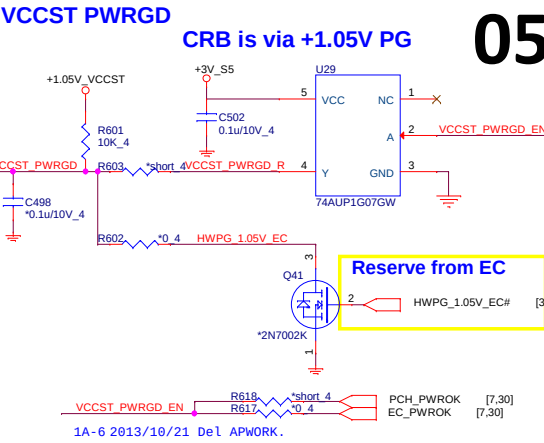
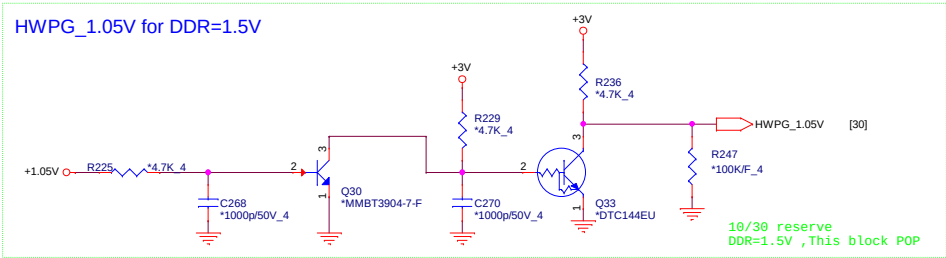
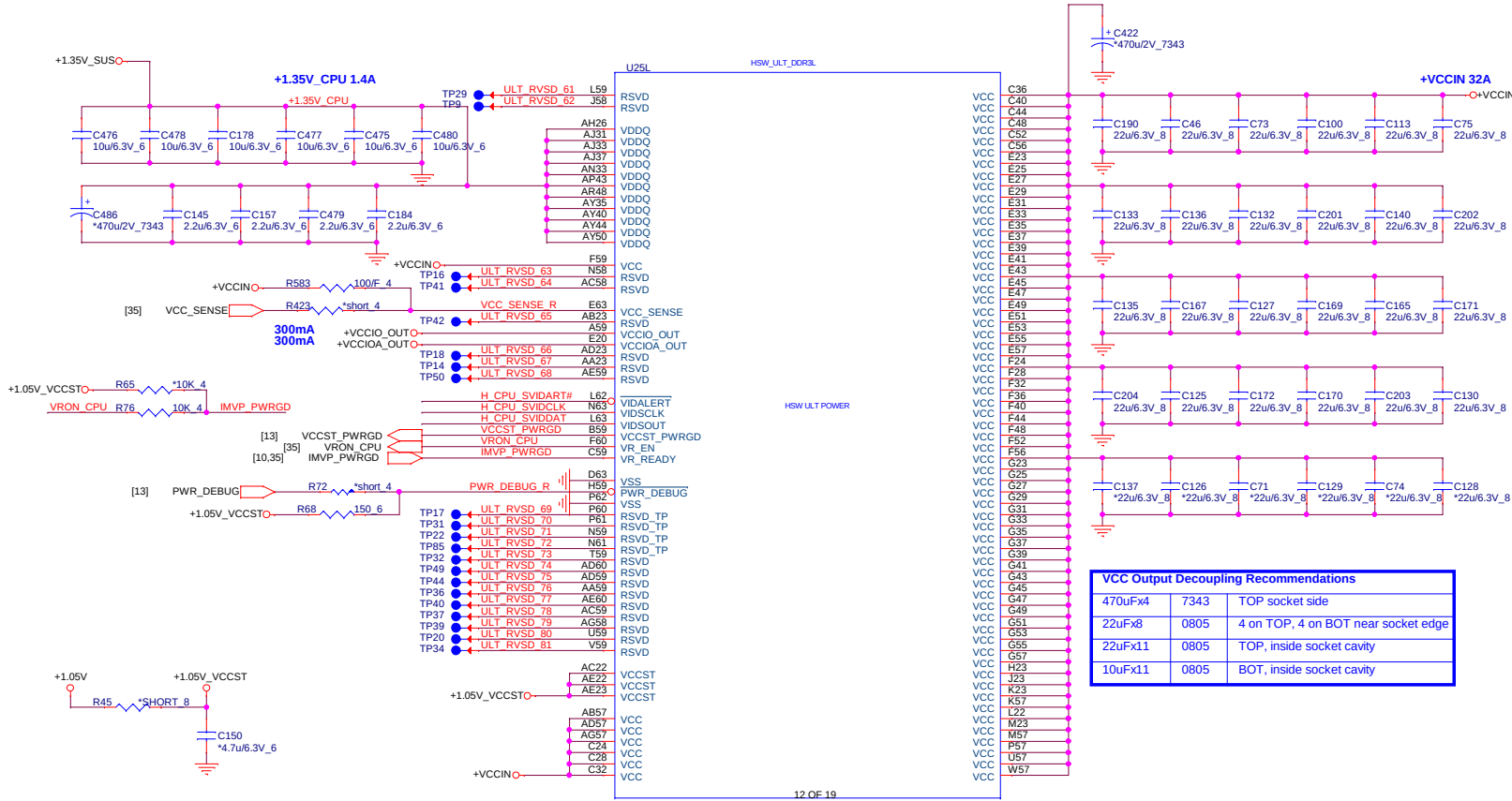


Quanta Computer Inc.
PROJECT : ZQ0

Size	Document Number Haswell 1/5 (PEG/DMI/FDI)	Rev 3A
Date:	Tuesday, April 08, 2014	Sheet 4 of 46

VDDQ Output Decoupling Recommendations				
330uFx2	7343	BOT socket side		
22uFx11	0805	5 on TOP, 6 on BOT inside socket cavity		
10uFx10	0805	5 on TOP, 5 on BOT inside socket cavity		

Haswell ULT (POWER)





	1	0	
CFG0 EAR-STALL/NOT STALL RESET SEQUENCE AFTER PCU PLL IS LOCKED	(DEFAULT) NORMAL OPERATION; NO STALL	STALL	
CFG1 PCH/ PCH LESS MODE SELECTION	(DEFAULT) NORMAL OPERATION	PCH-LESS MODE	
CFG3 PHYSICAL_DEBUG_ENABLED (DFX PRIVACY)	DISABLED NO PHYSICAL DISPLAY PORT ATTACHED TO EMBEDDED DISPLAY PORT	ENABLED AN EXTERNAL DISPLAY PORT DEVICE IS CONNECTED TO THE EMBEDDED DISPLAY PORT	
CFG 8 ALLOW THE USE OF NOA ON LOCKED UNITS	DISABLED(DEFAULT); IN THIS CASE, NOA WILL BE DISABLED IN LOCKED UNITS AND ENABLED IN UN-LOCKED UNITS	ENABLED; NOA WILL BE AVAILABLE REGARDLESS OF THE LOCKING OF THE UNIT	
CFG9 NO SVID PROTOCOL CAPABLE VR CONNECTED	VRS SUPPORTING SVID PROTOCOL ARE PRESENT	NO VR SUPPORTING SVID IS PRESENT. THE CHIP WILL NOT GENERATE (OR RESPOND TO) SVID ACTIVITY	
CFG10 SAFE MODE BOOT	POWER FEATURES ACTIVATED DURING RESET	POWER FEATURES (ESPECIALLY CLOCK GATINE ARE NOT ACTIVATED	



Quanta Computer Inc.
PROJECT : ZQ0

PROJECT : ZQ0

Size	Document Number	Rev
	Haswell 5/5 (CFG/GND)	3A
Date:	Monday, April 07, 2014	Sheet 6 of 46

07

Power Sequence

[30] 
Speed up 250ms to boot up
for EC power on 250 ms

PCH PM PU/PD

PLTRST# Buffer

SYSPWOK

Deep Sx Circuit

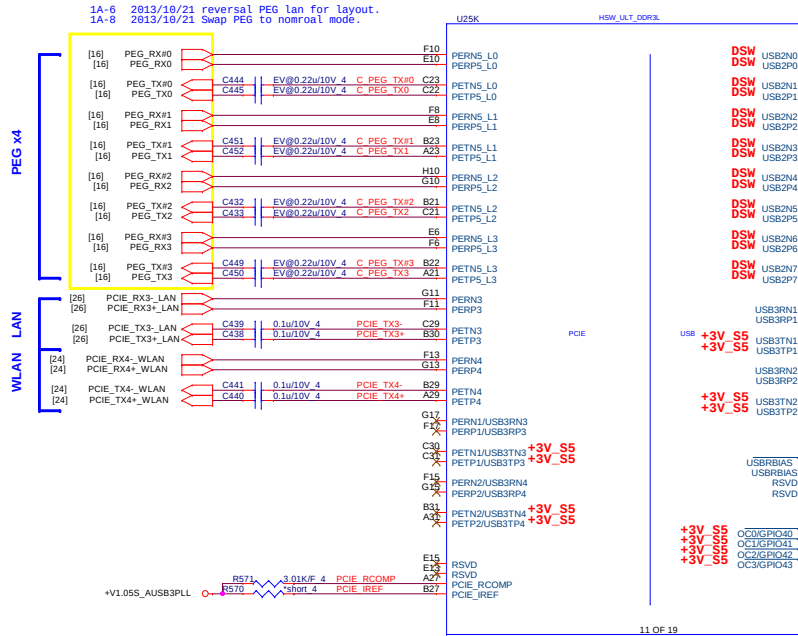


Quanta Computer Inc.
PROJECT : ZQ0

Size	Document Number LPT 1/6 (DMI/FDI/VGA)	Rev 3A
Date:	Tuesday, April 08, 2014	Sheet 7 of 46

Haswell ULT PCH (PCIE,USB3.0,USB2.0)

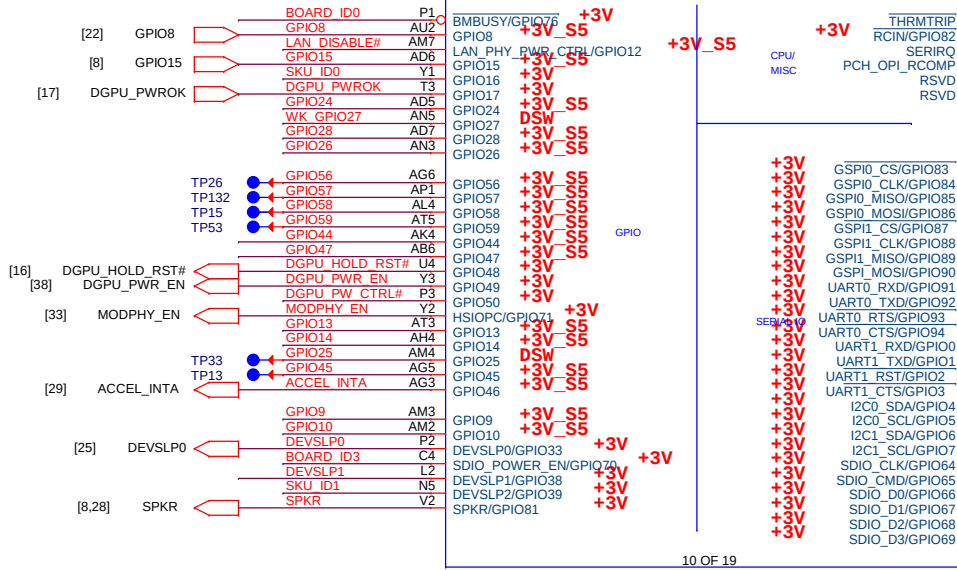
1A-6 2013/10/21 reversal PEG lan for layout.
1A-8 2013/10/21 Swap PEG to nomral mode.



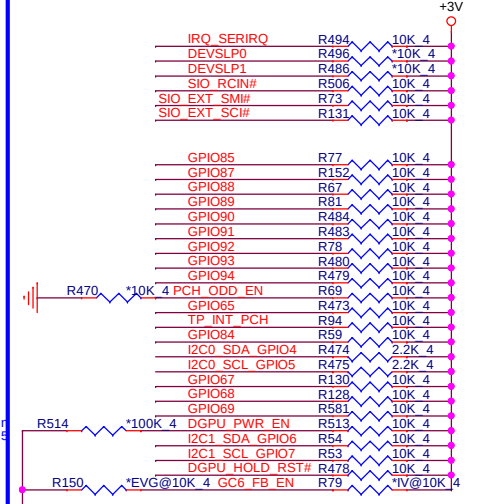
Haswell ULT PCH (GPIO, CPU/MISC, NCTF)

10

	High	Low
GPIO8	Touch panel	No touch panel



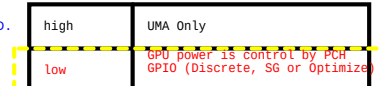
PCH GPIO PU/PD



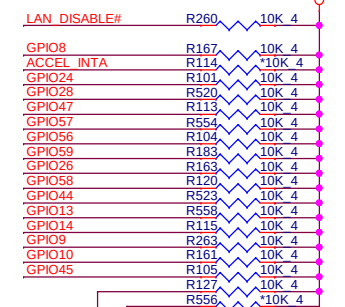
1A-1 20131015 For GC6 NV DG GC6_FB_EN PD.

1A-8 20131022 Change GPIO83/84 GPU GC6 pin to GPIO2/3.

1A-12 20131029 Change GPIO45 to PU S5, duple GPIO58 one is GPIO56.



DGPU_PWROK PD on GPU side



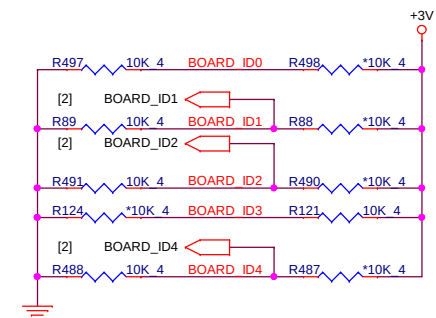
1B-7 20131220 Change +3VPCU to +3V_S5 non deep sx
GPIO27 : If not used then use 8.2-k Ω to 10-k Ω pull-down to GND.

Quanta Computer Inc.
PROJECT : ZQ0

Size	Document Number	Rev
	LPT 4/6 (GPIO/MISC)	3A

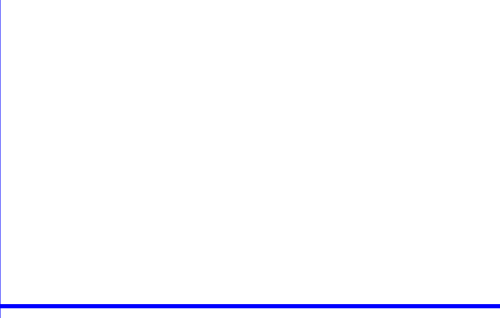
Date: Tuesday, April 08, 2014 Sheet 10 of 46

Board ID



	Low	High
BOARD_ID0	N15V-GL-B	N15V-GM-B
BOARD_ID1	Reserved (Default)	Reserve
BOARD_ID2	Reserve for Touch pad, default(low)	
BOARD_ID3	DTPM	No DTPM
BOARD_ID4	Non-Dolly (Default)	Dolly

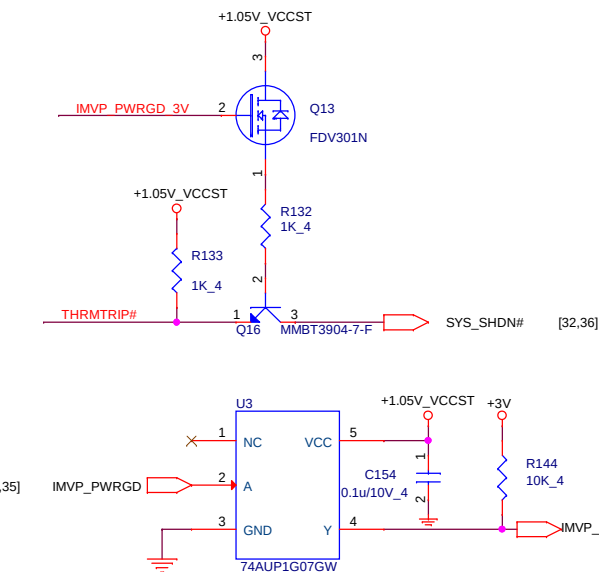
RAM ID

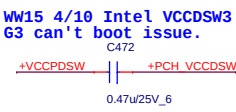


SKU ID

	SKU_ID1	SKU_ID0	VGA H/W Signal	Setup Menu	UMA boot
UMA Only	0	0	UMA	Hidden	UMA boot
dGPU Only	0	1	GPU	Hidden	GPU boot
Switchable (Mux)	1	0	UMA+GPU	dGPU/SG	UMA boot
Optimize (Muxless)	1	1	UMA	UMA/SG	UMA boot

CPU thermal trip





1A-1 2013/10/11 del LDO change to MOS.



11mA

+3V_S5

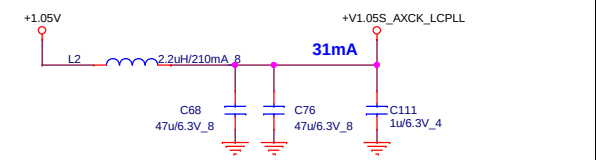
+V3.3DX_1.5DX_1.8DX_AUDIO

R196

*Short 6

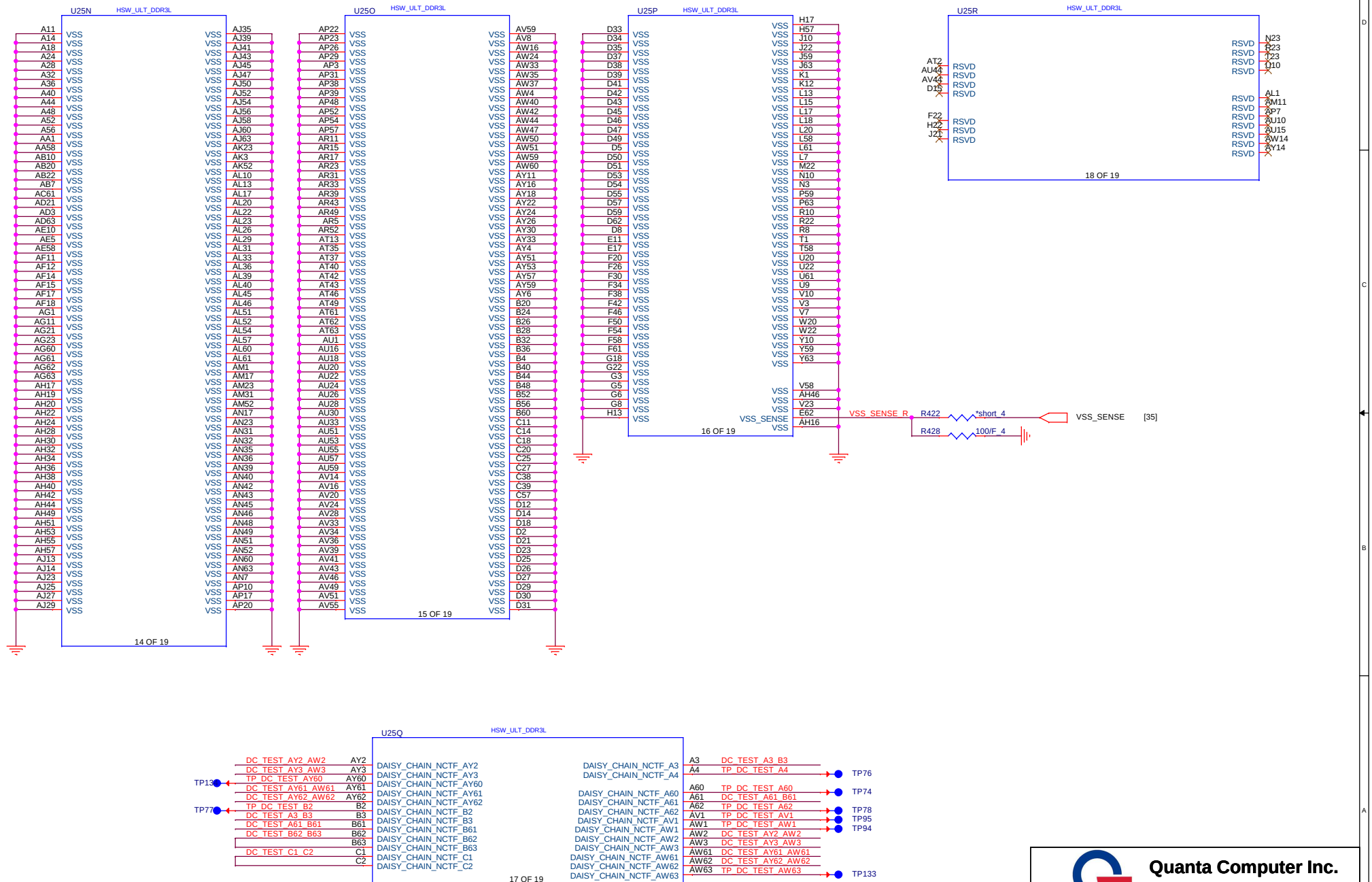
C155
0.1uF10V_4

Place close to bal

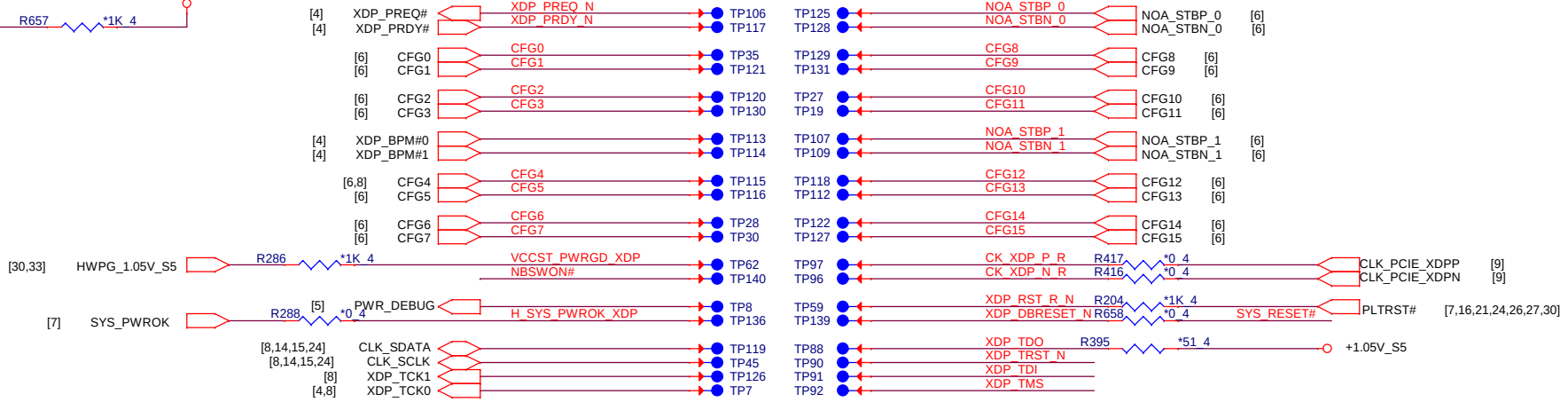
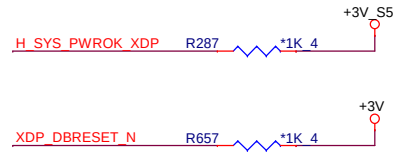


 Quanta Computer Inc. PROJECT : ZQ0	
Size	Document Number LPT 5/6 (POWER)
Date	Monday, April 07, 2014
Sheet	11 of 46

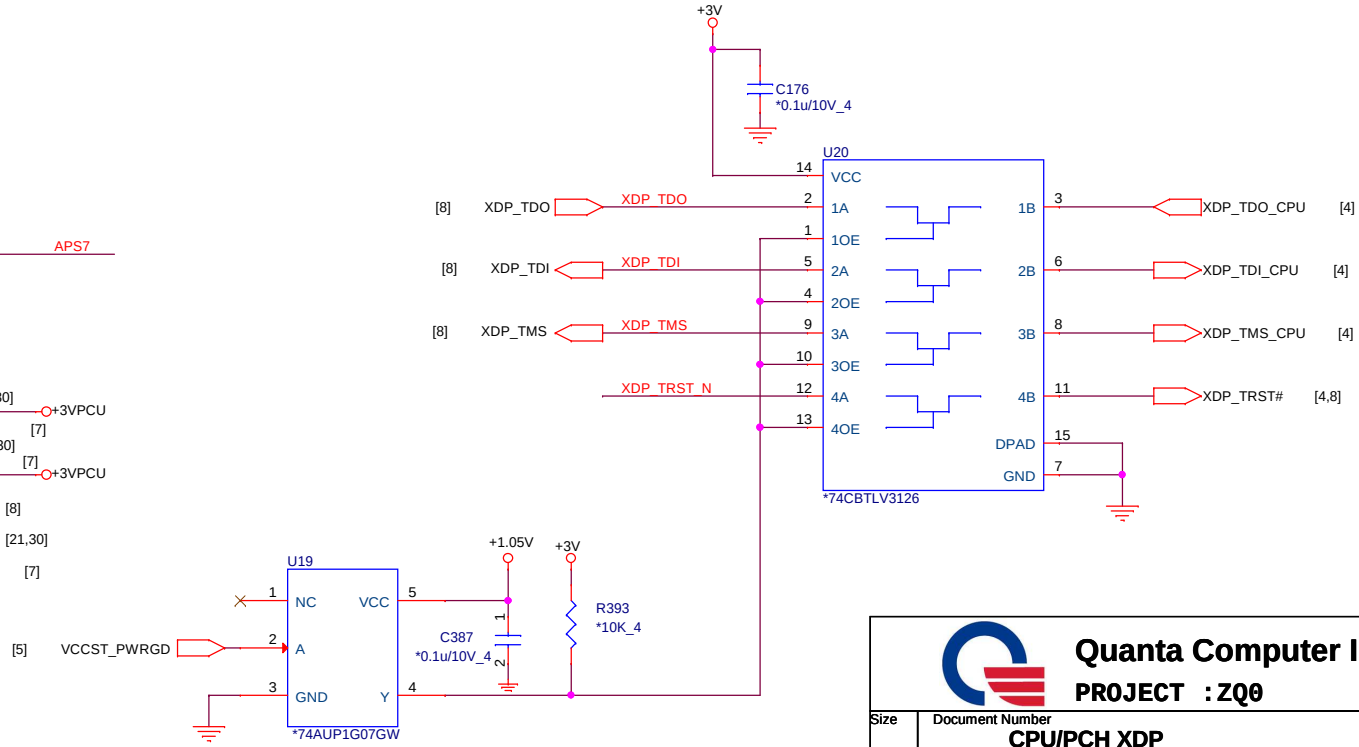
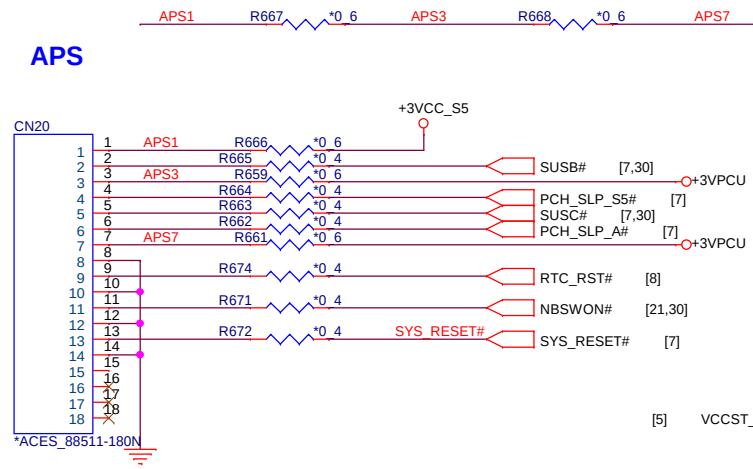
Haswell ULT (GND)

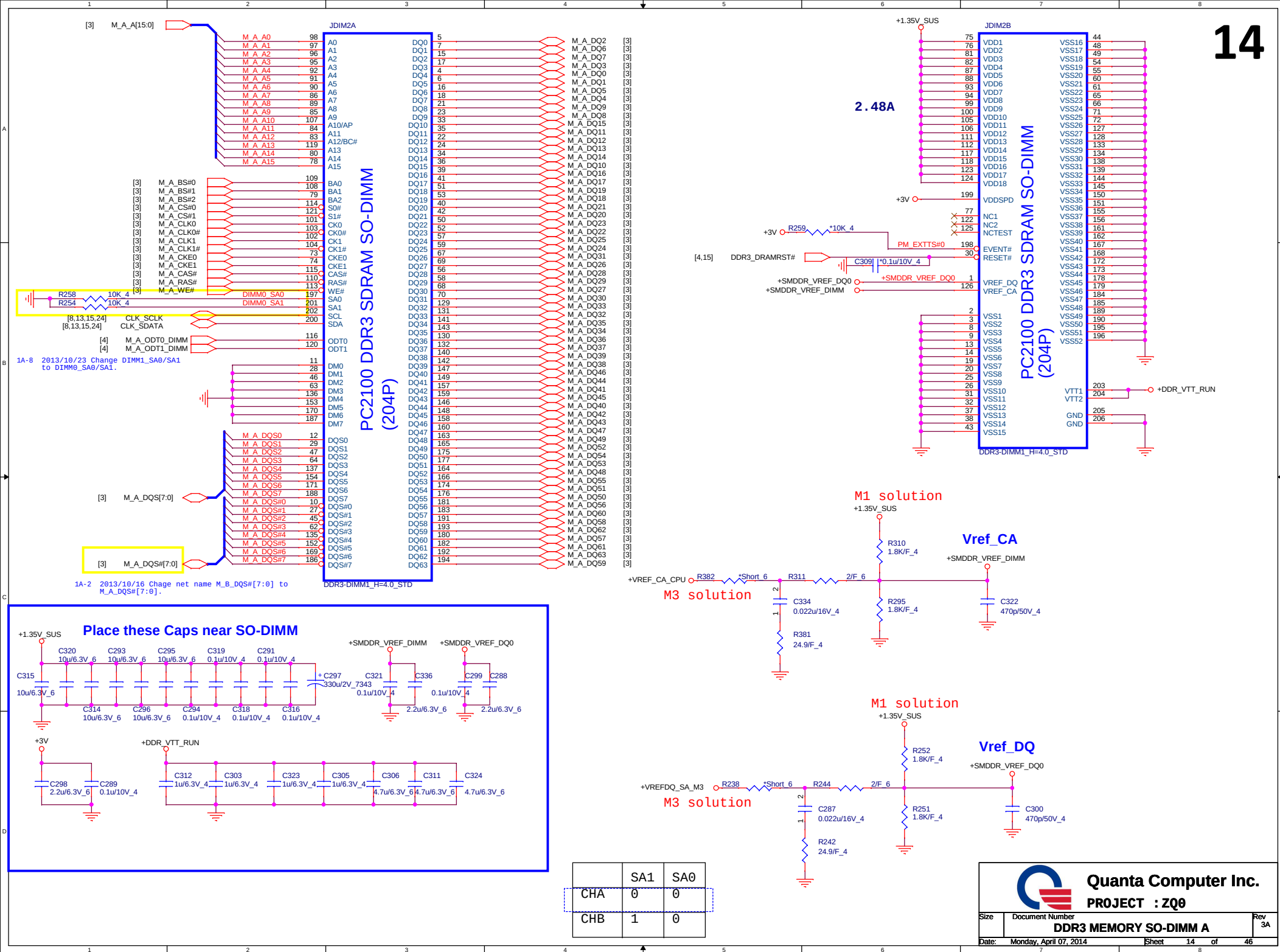


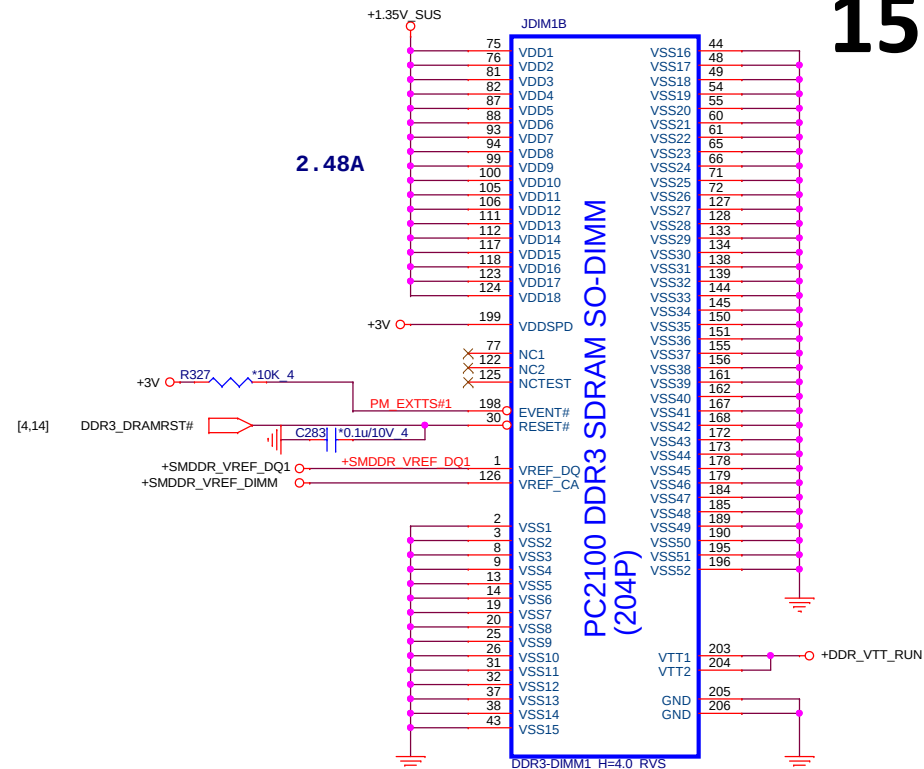
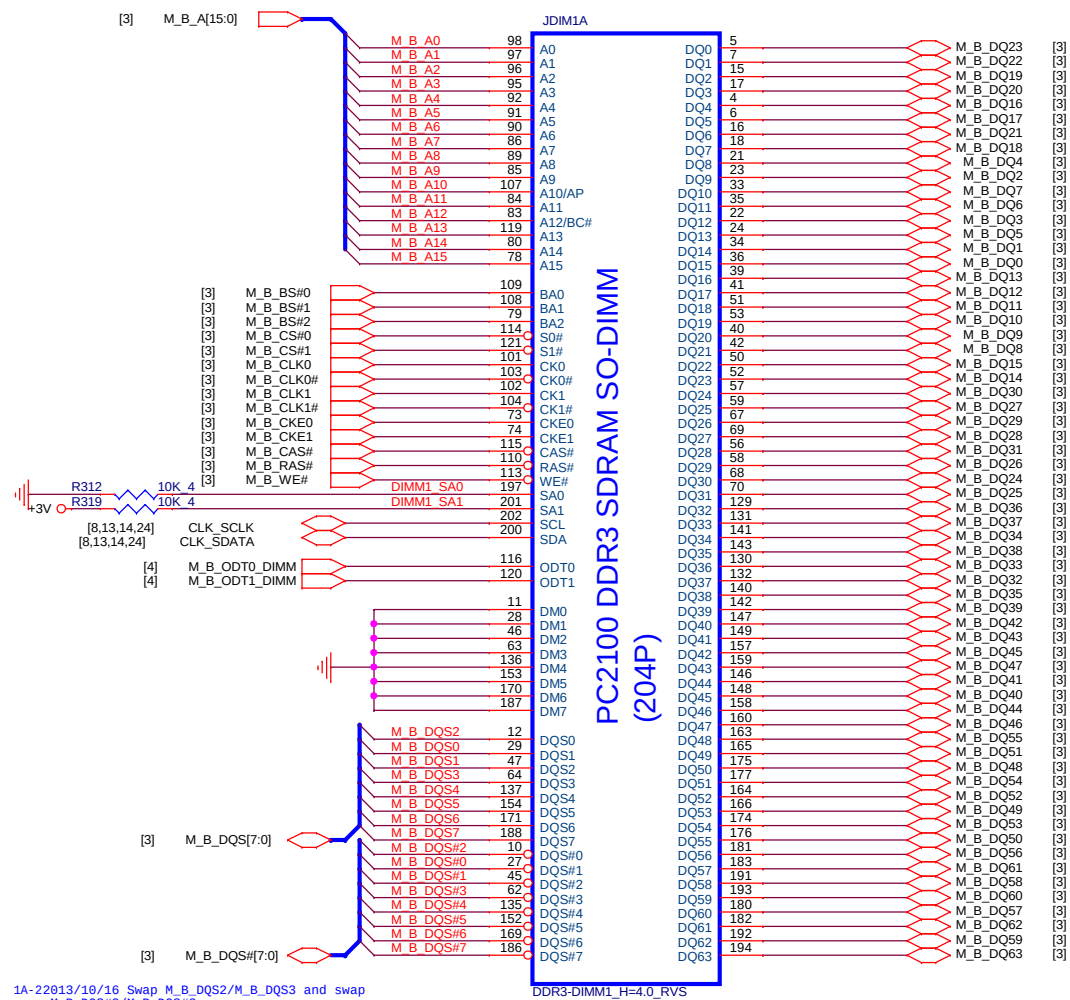
Quanta Computer Inc.
PROJECT : ZQ0



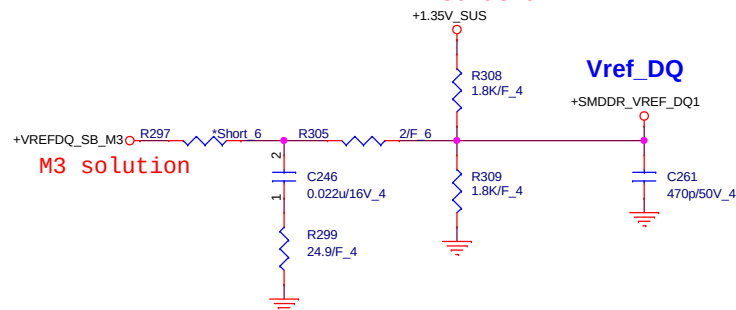
APS



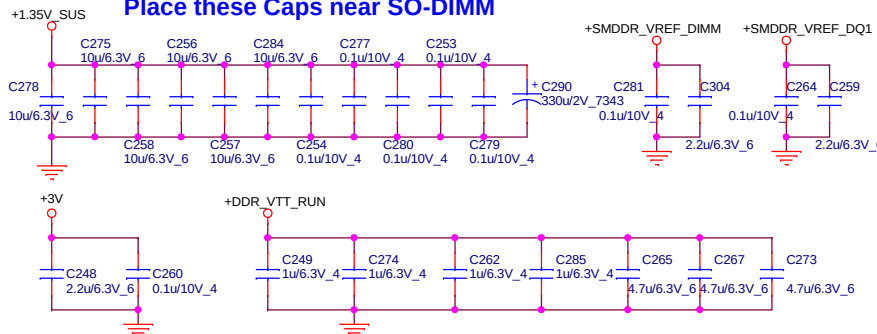




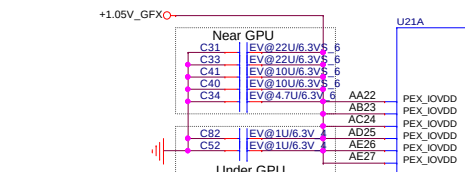
M1 solution



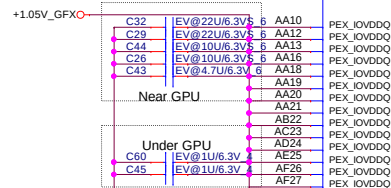
Place these Caps near SO-DIMM



	SA1	SA0
CHA	0	0
CHB	1	0



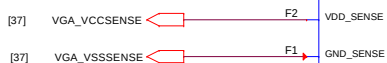
PEX_IOVDD + PEX_IOVDDQ = 1.042A



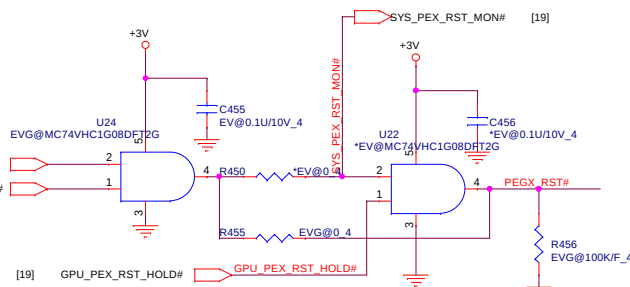
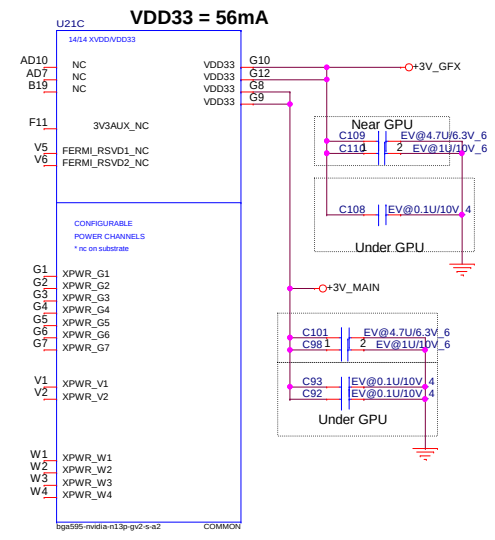
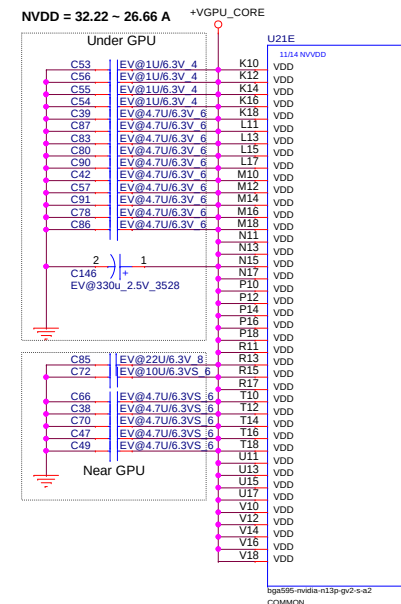
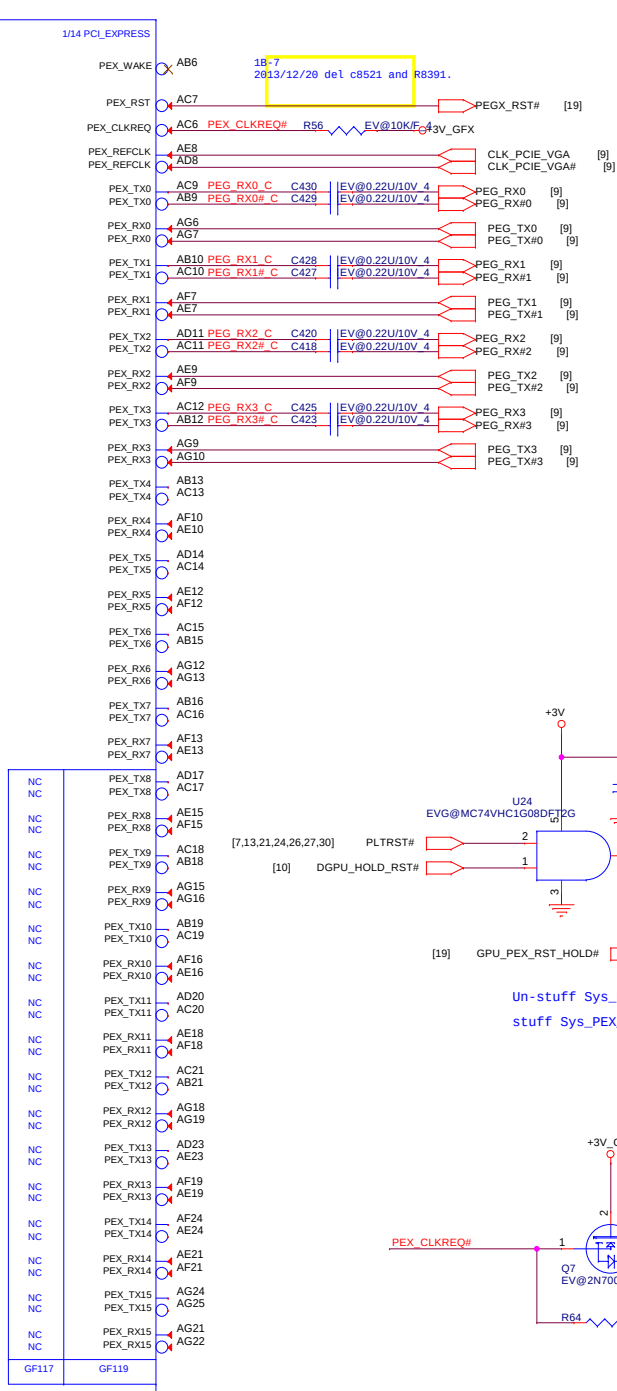
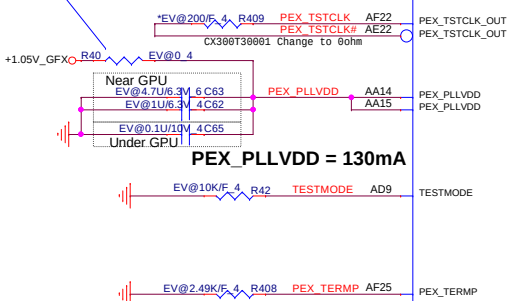
PEX_PLL_HVDD + PEX_SVDD_3V3 = 143mA



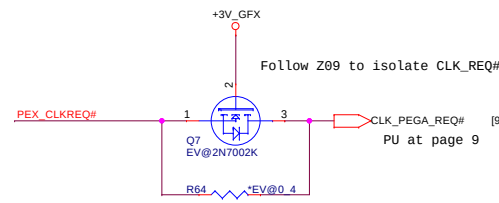
8mils width (0.2MM)



1B-5 2013/12/17 Change R8051 to 0402 size.

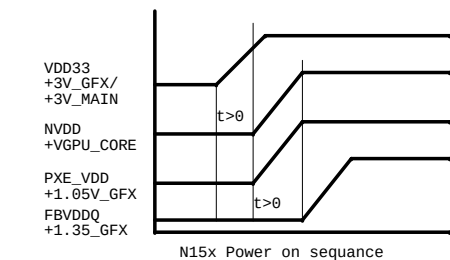


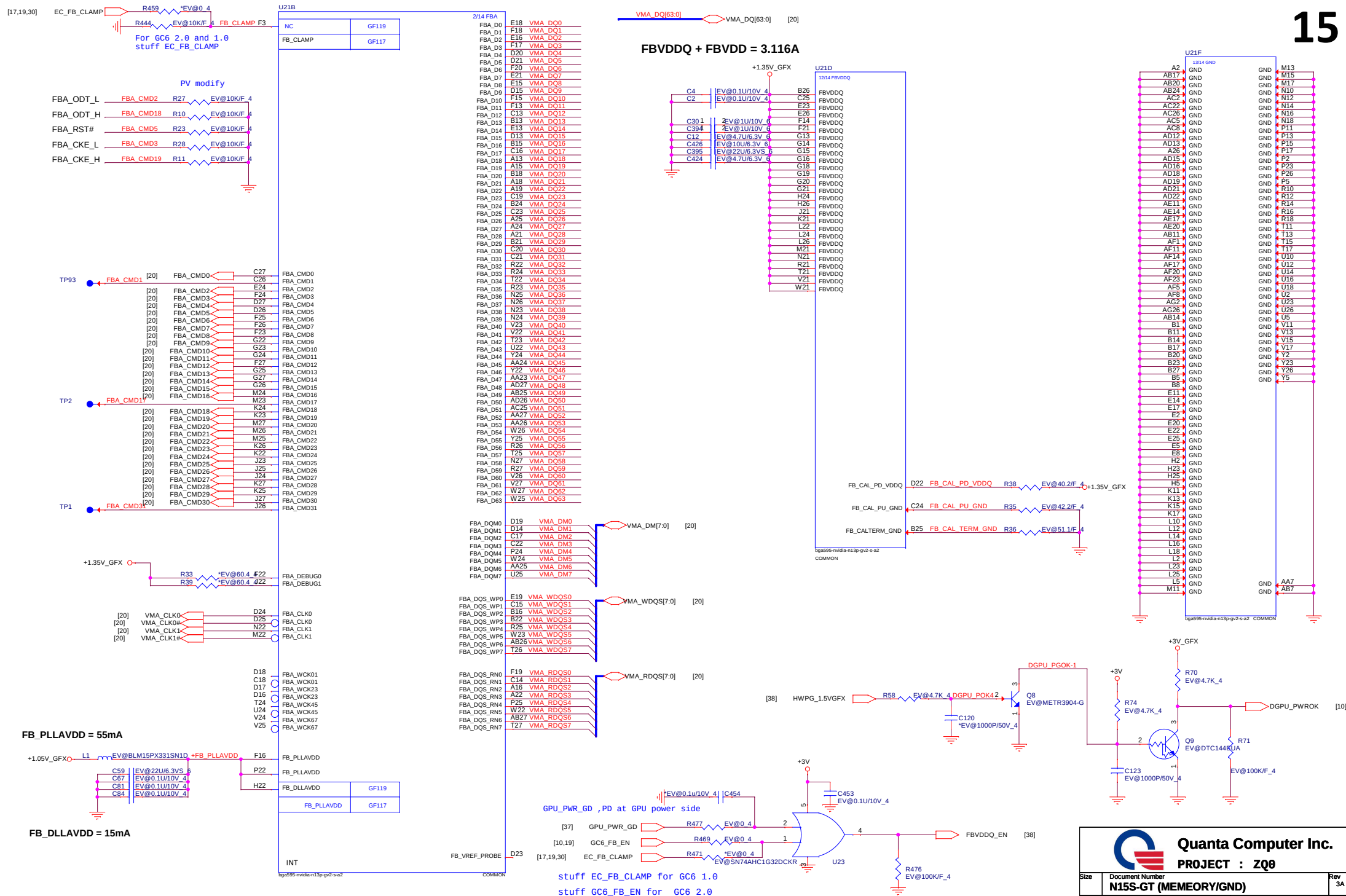
Un-stuff Sys_PEX_RST_MON# , stuff PEGX_RST# for not GC6
stuff Sys_PEX_RST_MON# for GC6

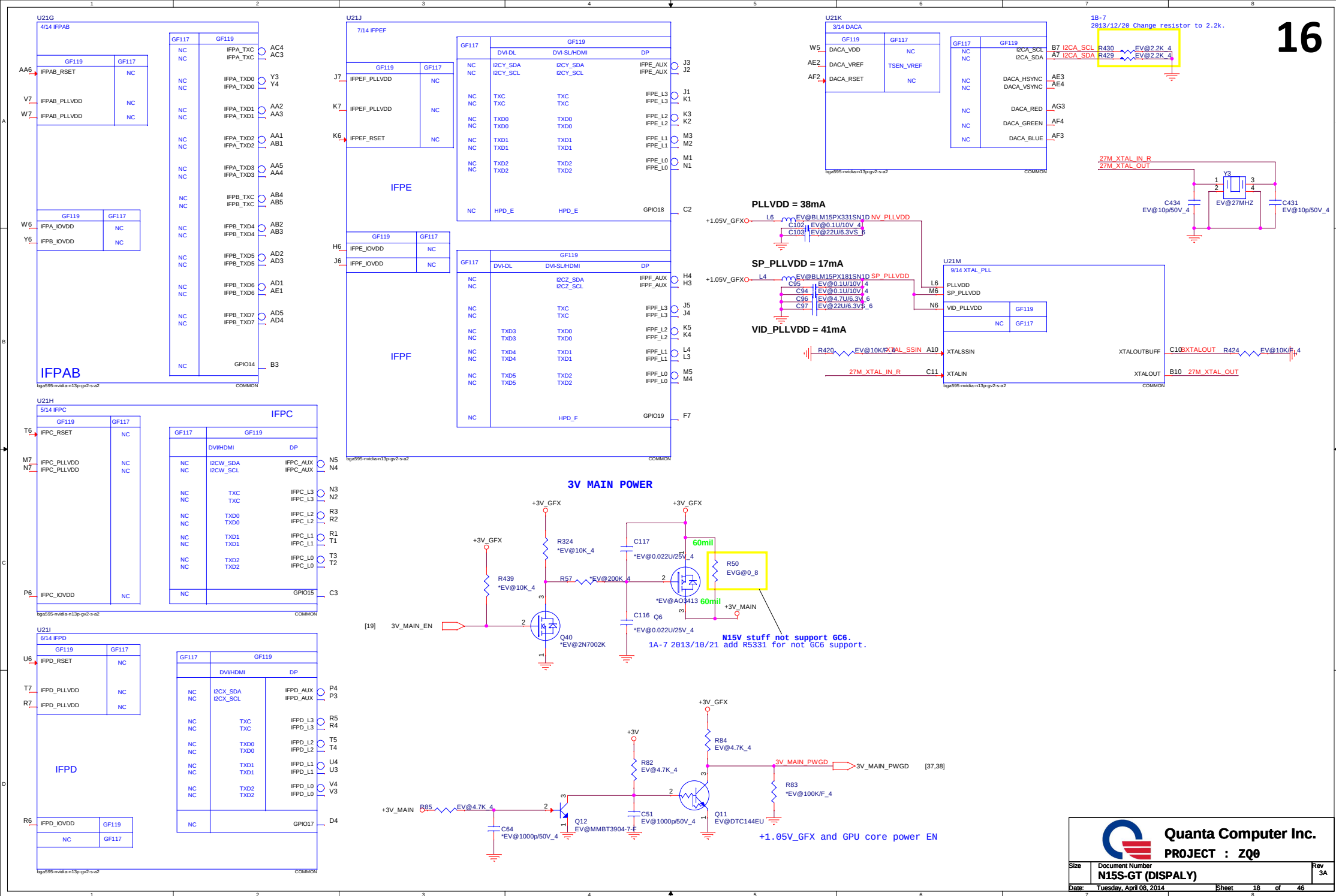


Power down sequence

First Rail to Power Down
Last Rail to Power Down



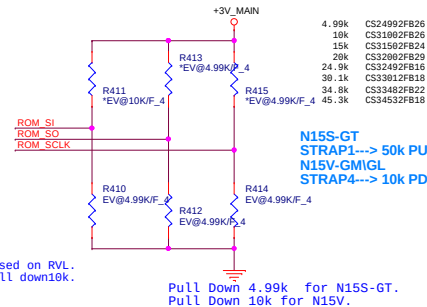
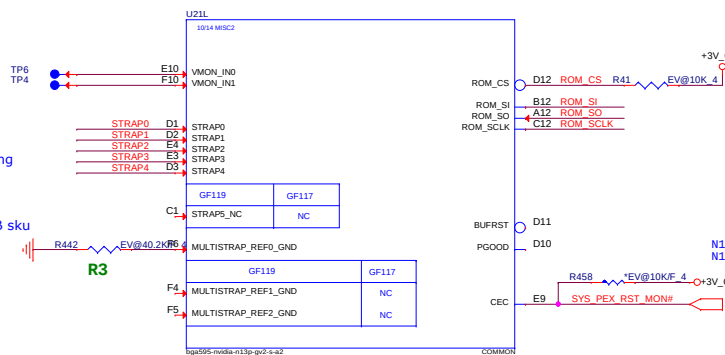




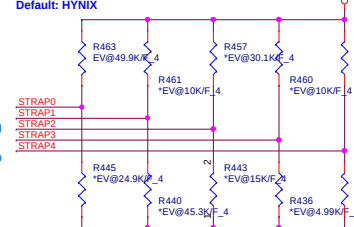
For N15S-GT sku
N15S-GT device ID=0x0FE4
R3=40.3k pull down.

- 1.ROM_SCLK = 4.99K pull down
- 2.ROM_SO = 4.99K pull down
- 3.ROM_SI= Memory strap setting
- 3.STRAP0 = 50k Pull pu.
- 4.Strap4~1 = reserve Pull pu and Pull down

For N15V-GL-B and N15V-GM-B sku
Board_ID0=
H=N15V-GM,L=N15V-GL
Device ID=0x1140
R3= N.C.
1.ROM_SCLK=10K pull down.
2.ROM_SI= 10k pull down
3.ROM_SO= 10k pull down
4.Strap3~0 = RVL memory
binary mode setting.
5.Strap4 =10k pull down

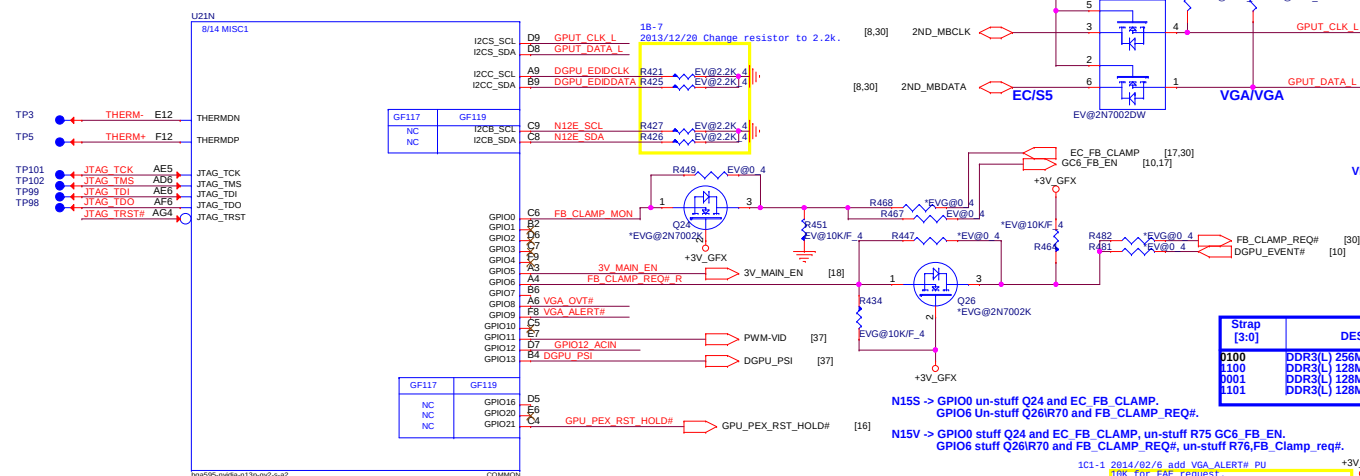


Default: HYNIX



N15S Strap0 pull up 50k, strap1~4 reverse only.
N15V Strap4 pull down 10k, strap0~3 based on RVL binary setting.

	PU-VDD	PD
4.99K	1000	0000
10K	1001	0001
15K	1010	0010
20K	1011	0011
24.9K	1100	0100
30.1K	1101	0101
34.8K	1110	0110
45.3K	1111	0111



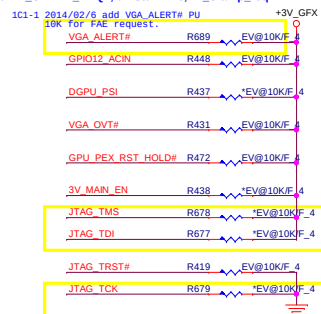
VRAM Configuration Table

ROM_SI	DESCRIPTION	Vendor	Vendor PIN	QCI PIN	STN PIN
0000	DDR3(L) 256MBx16x4, 64bit,1000MHz	HYNIX	H5TCA46G3AFR-11C	AKD5PGWTW05	AKD5PGWTW1
0100(0101)	DDR3(L) 256MBx16x4, 64bit,1000MHz(900MHz)	SAMSUNG	K4W461648F0-BC1A		
0110(1003)	DDR3(L) 256MBx16x4, 64bit,1000MHz(900MHz)	HYNIX	H5TCA26G3AFR-11C		
0111(1010)	DDR3(L) 128MBx16x4, 64bit,1000MHz(900MHz)	MICRO	MT41J256M1G31-0993G:K		
1000(1011)	DDR3(L) 128MBx16x4, 64bit,1000MHz(900MHz)	SAMSUNG	K4W42616460-BC1A		
0001(0100)	DDR3(L) 128MBx16x4, 64bit,1000MHz(900MHz)	MICRO	MT41J256M1G31A-0936: E		

Strap [3:0]	DESCRIPTION	Vendor	Vendor P/N	QCI P/N	Note
0100	DDR3(L) 256MBx16x4, 64bit,1000MHz(900MHz)	HYNIX	H5TCA4663AFR-11C		
1101	DDR3(L) 128MBx16x4, 64bit,1000MHz(900MHz)	HYNIX	H5TCA4663AFR-11C		
0001	DDR3(L) 128MBx16x4, 64bit,1000MHz(900MHz)	MICRO	MT41J128M16JT-093G:K		
1101	DDR3(L) 128MBx16x4, 64bit,1000MHz(900MHz)	MICRO	MT41J256M16HA-093G:E		

N15S -> GPIO0 un-stuff Q24 and EC_FB_CLAMP.
GPIO6 Un-stuff Q26R70 and FB_CLAMP_REQ#

N15V -> GPIO0 stuff Q24 and EC_FB_CLAMP, un-stuff R75 GC6_FB_EN.
GPIO6 stuff Q26/R70 and FB_CLAMP REO#. un-stuff R76.FB_Clamp req#



1C-2 2014/01/13 add R678\R677 PU and R679 PD
for ICT.

GPIO	I/O	PIN	USAGE
0	IN	FB_CLAMP_MON	FB Clamp monitor
1	OUT	MEM_VDD_CTL	Memory VDD VID
2	OUT	LCD_BL_PWM	Panel Backlight PWM
3	OUT	LCD_VCC	PANEL POWER ENABLE
4	OUT	LCD_BLEN	PANEL BACKLIGHT ENABLE
5	OUT	Reserved	--
6	OUT	FB_CLAMP_TGL_REQ	Active low FB Clamp toggle request
7	OUT	3D_VISION	3D VISION LEFT/RIGHT signal
8	I/O	OVERT	ACTIVE LOW THERMAL OVER TEMP
9	I/O	ALERT	ACTIVE LOW THERMAL ALERT
10	OUT	MEM_VREF_CTL	MEMORY_VREF CONTROL
11	OUT	PWR_VID	GPU_CORE_VDD PWM Control signal
12	IN	PWR_LEVEL	AC Power detect or power supply overdraw input
13	OUT	PSI	Phase Shedding

GPIO ASSIGNMENTS



Quanta Computer Inc.
PROJECT : ZQ0

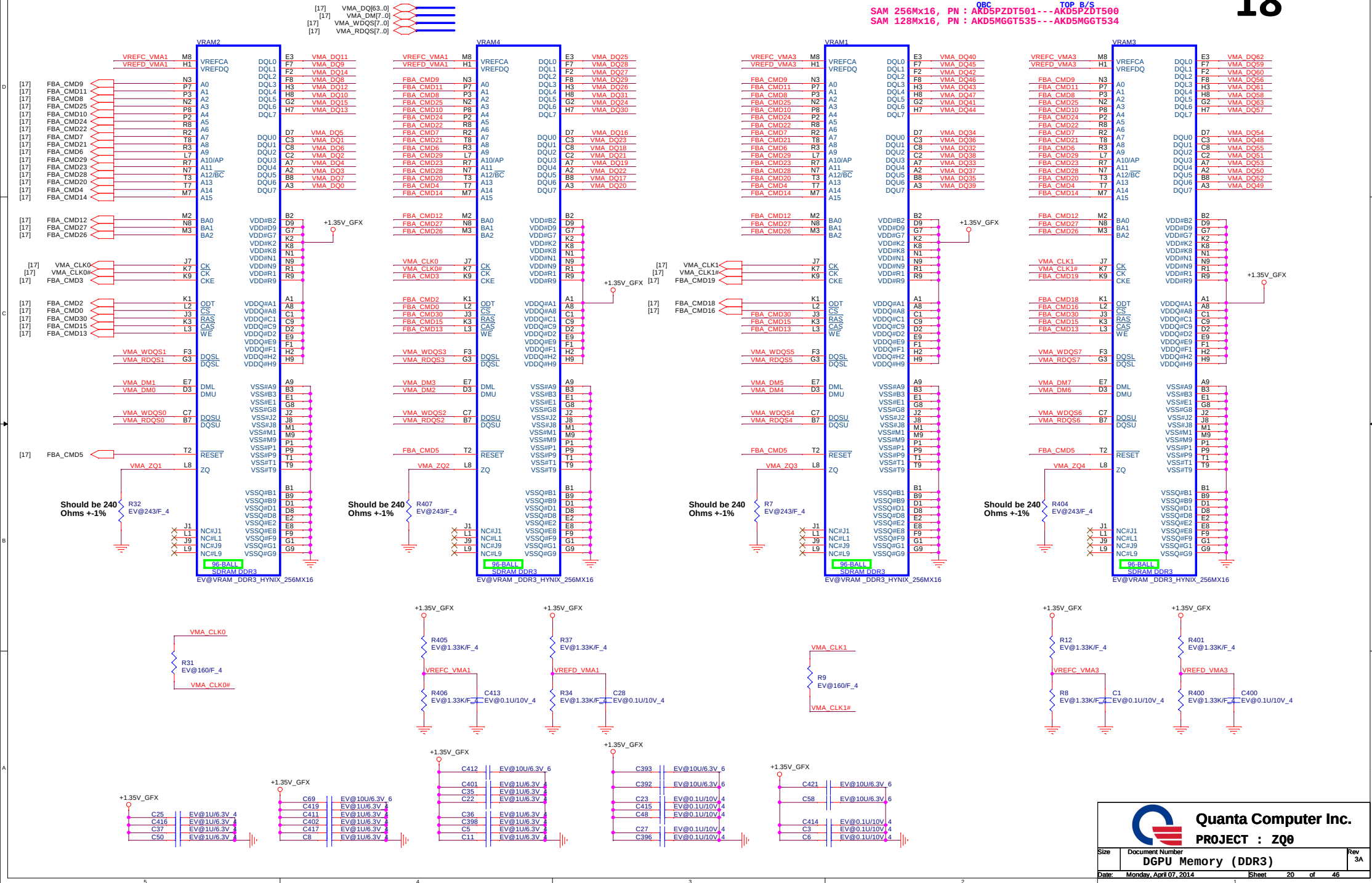
Size	Document Number N15S-GT (GPIO/STRAPS)	Rev 3A
Date	Monday, April 07, 2014	Sheet 19 of 46

CHANNEL A: 256MB/512MB DDR3

HYU 256Mx16, PN : AKD5PGWTW08 --- AKD5PGWTW07
HYU 128Mx16, PN : AKD5MZDTW03 --- AKD5MZDTW02

SAM 256Mx16, PN : AKD5PZDT501 --- AKD5PZDT500
SAM 128Mx16, PN : AKD5MGGT535 --- AKD5MGGT534

18

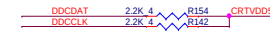
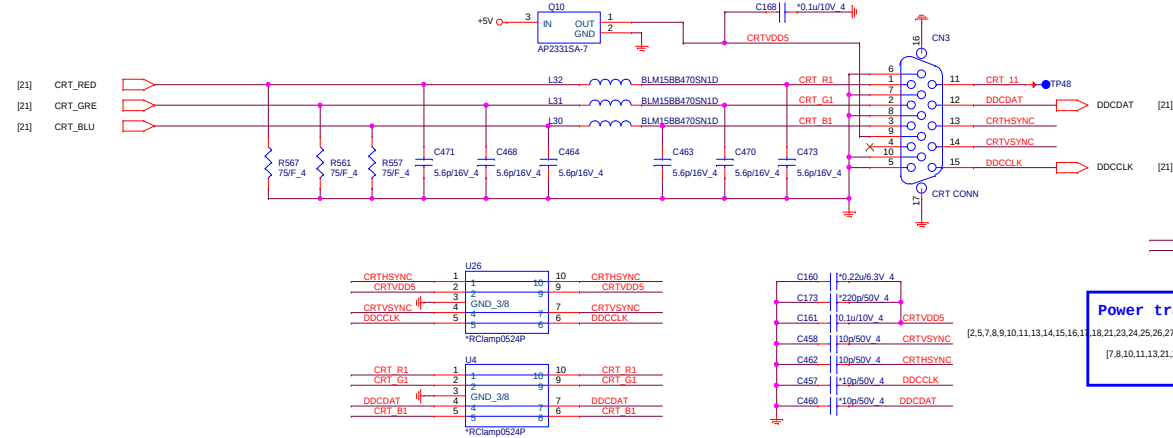
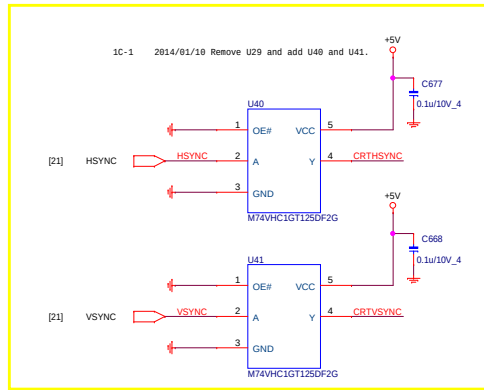


Quanta Computer Inc.
PROJECT : ZQ0

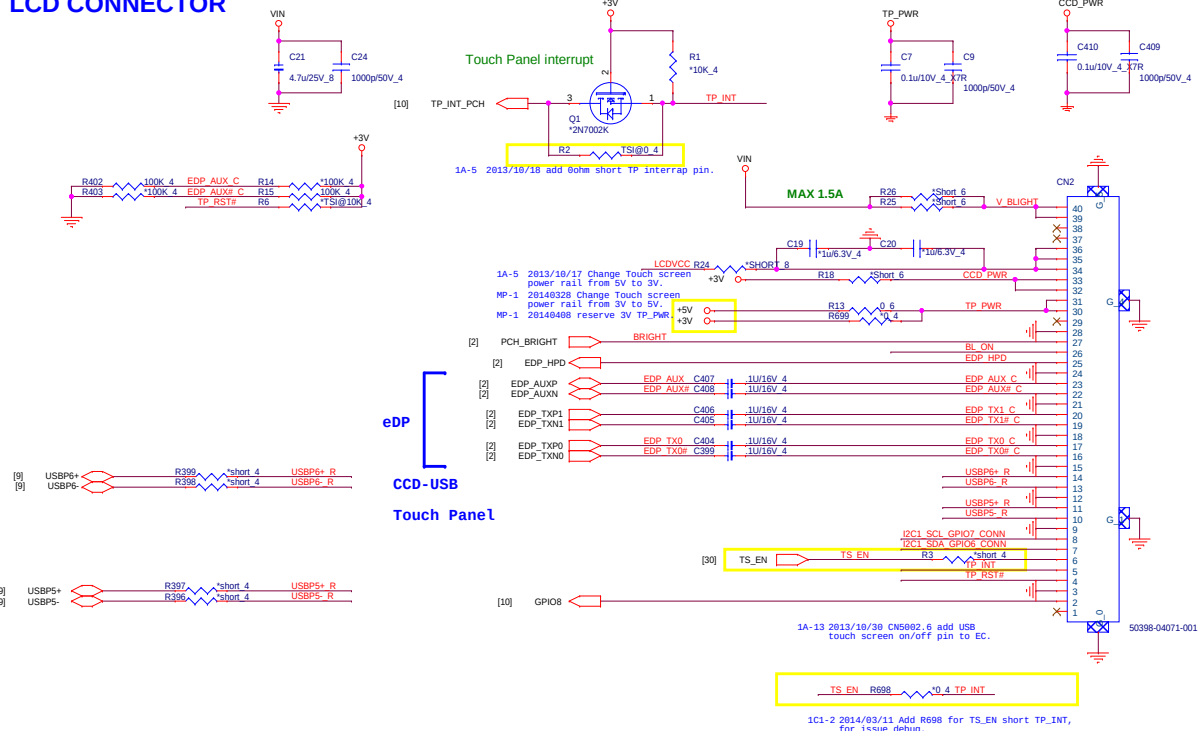
Size Document Number
DGPU Memory (DDR3)

Date: Monday, April 07, 2014 Sheet 20 of 46

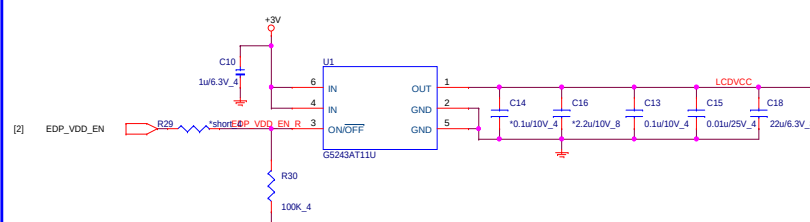
CRT



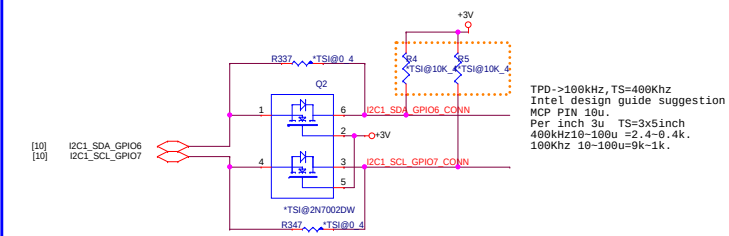
LCD CONNECTOR



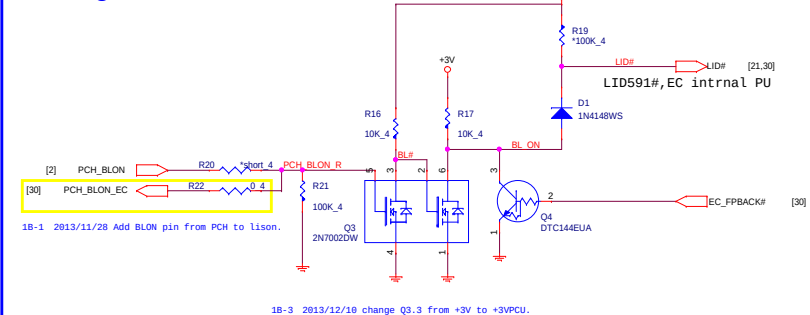
LCD Power



Touch screen level shift I2C(reserve)

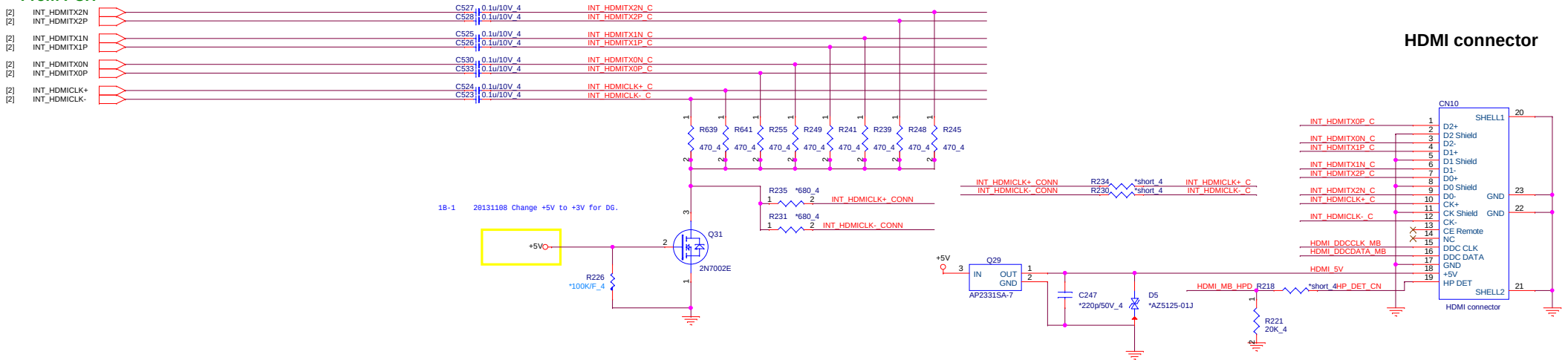


Backlight Control

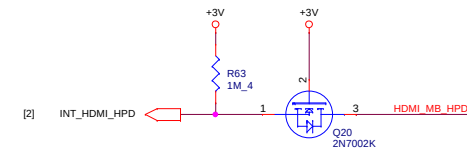


HDMI

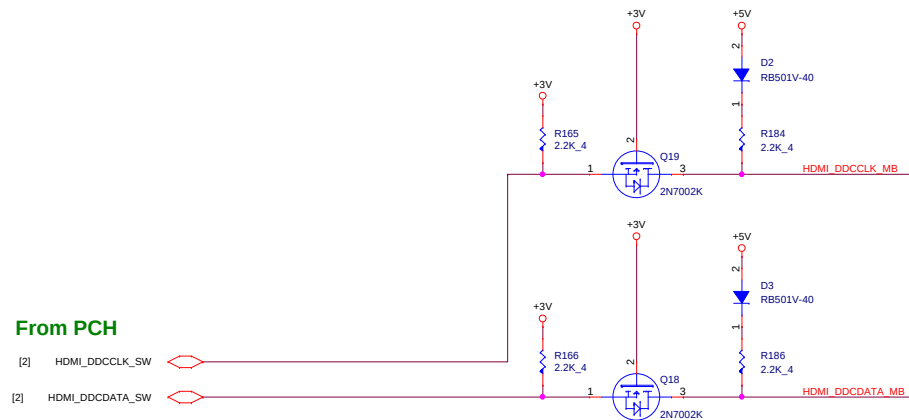
From PCH



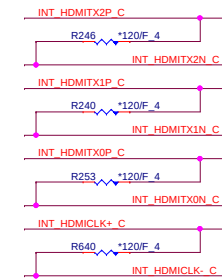
HDMI-detect



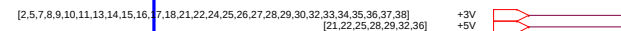
I2C

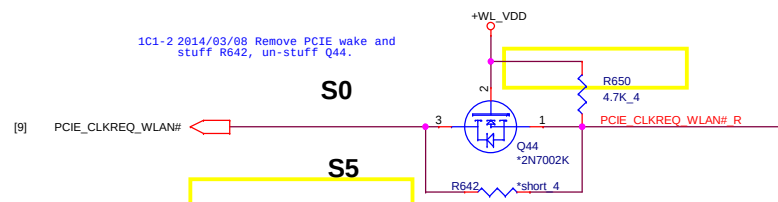
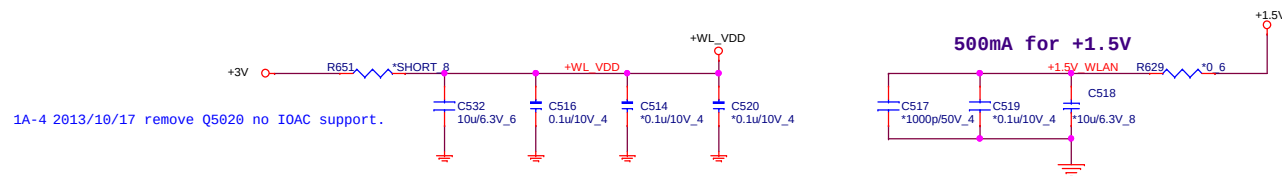
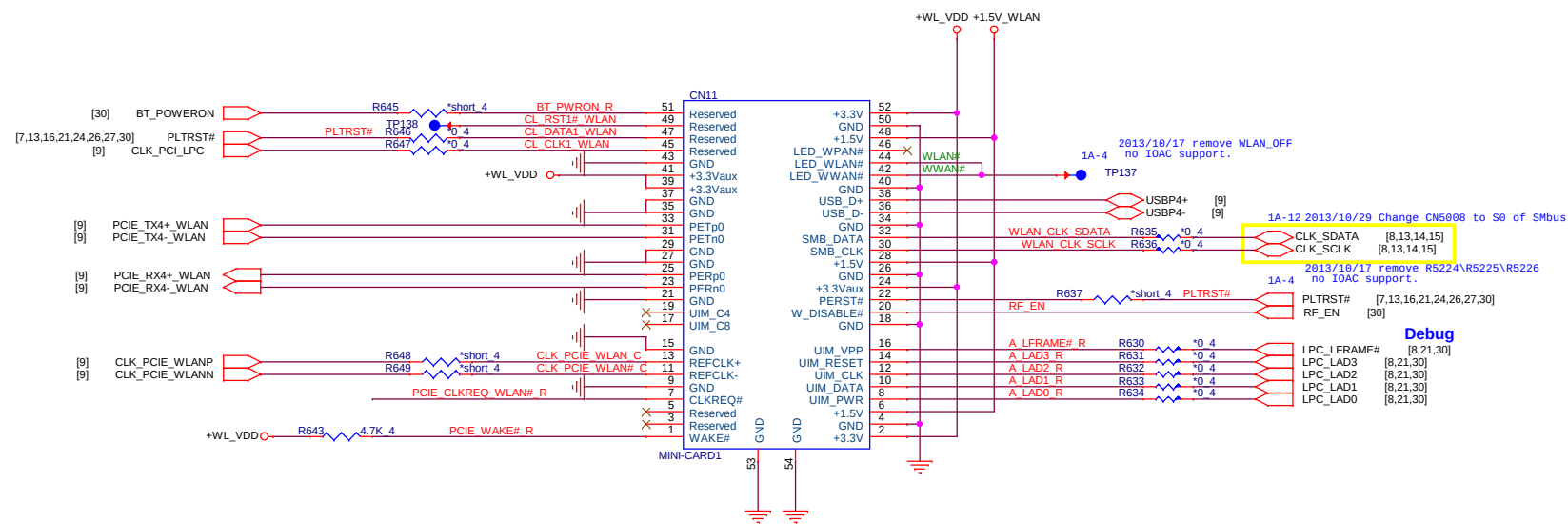


EMI

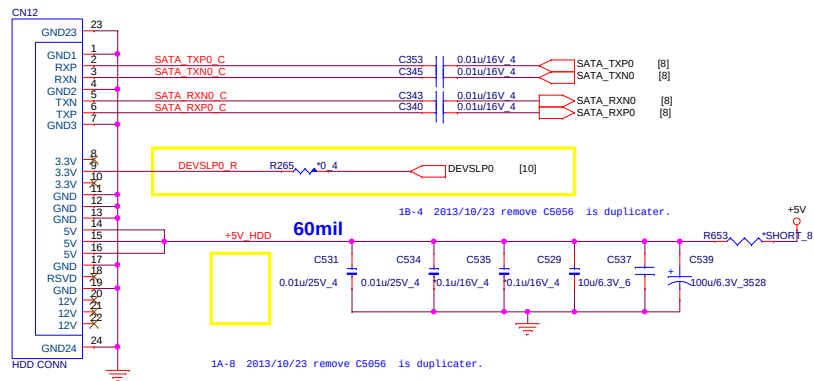


Power trace tracking



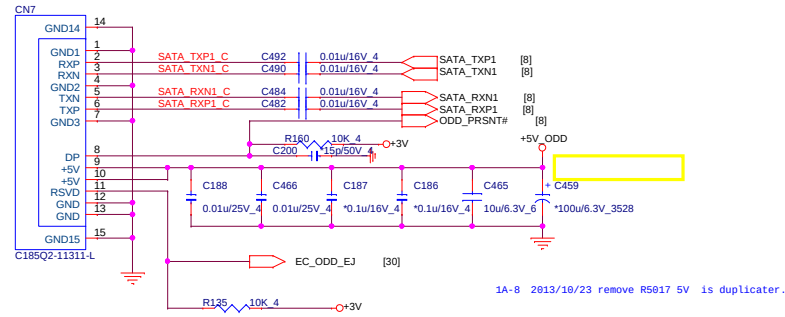


2.5" SATA HDD (HDD)

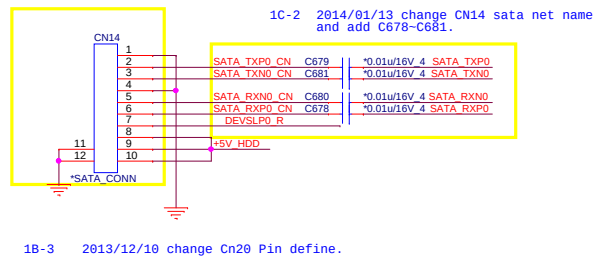


SATA ODD Connector

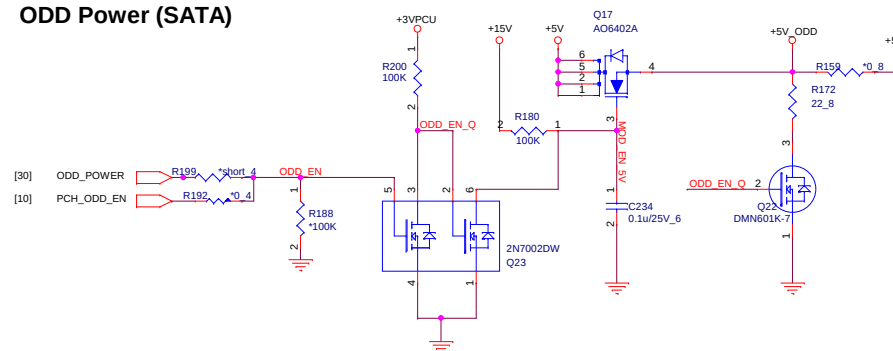
27



FFC Type SATA HDD CON



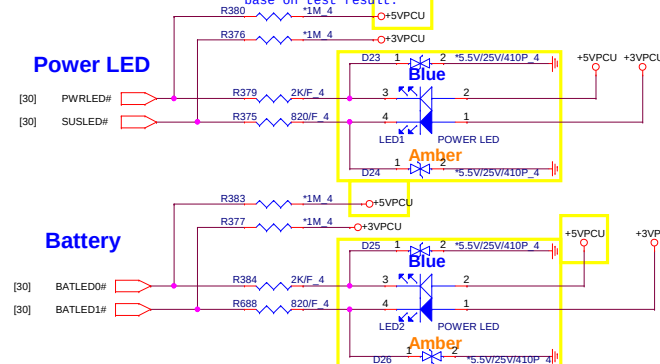
ODD Power (SATA)



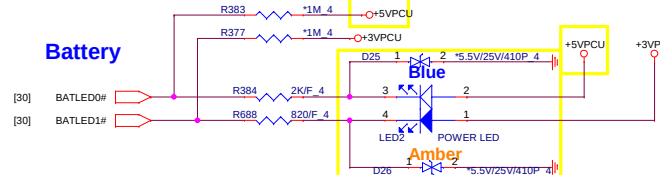
POWER LED

1A-9 2013/10/17 Change power LED from +3VPCU to +3V_S5.
 1A-10 2013/10/25 change LED from 3pin to 4pin.
 for acer request,
 1A-11 2013/10/28 change LED from 4pin to 3pin.
 and power LED to +3VPCU.
 1B-2 2013/12/03 change LED from 3pin to 4pin.
 1C1-1 2014/02/06 change Blue LED power rail to +5VPCU.
 1C1-1 2014/02/11 add ESD on led.
 1C1-1 2014/02/13 change LED to lite-on and resistor
 base on test result.

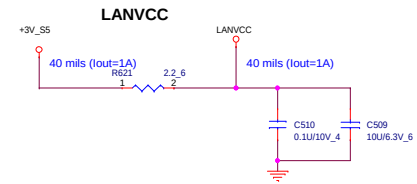
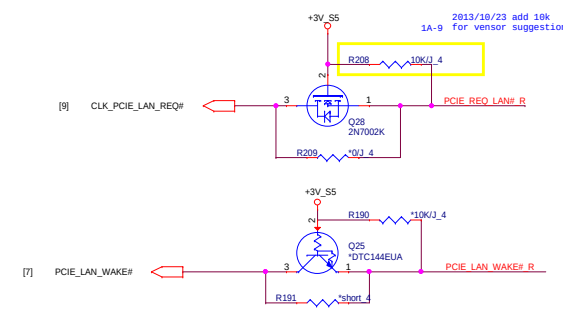
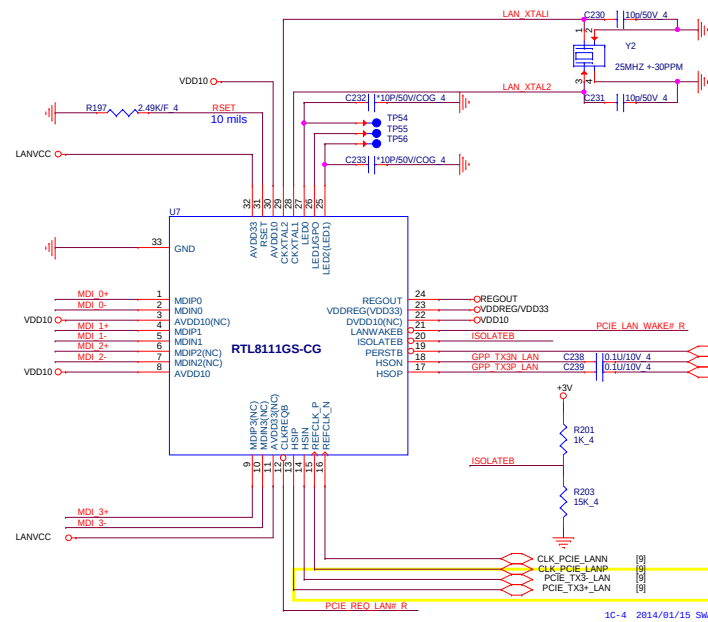
Power LED



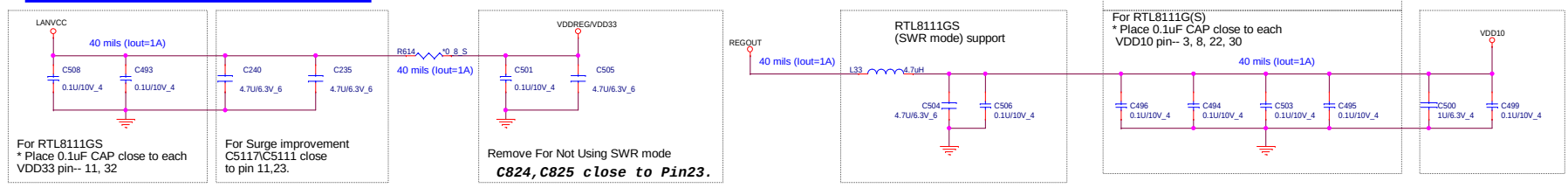
Battery



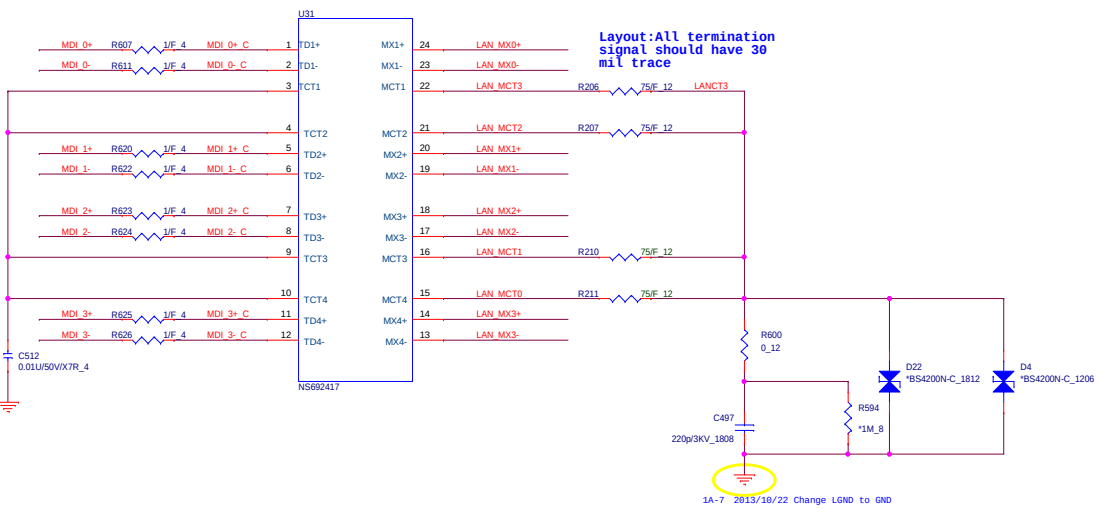
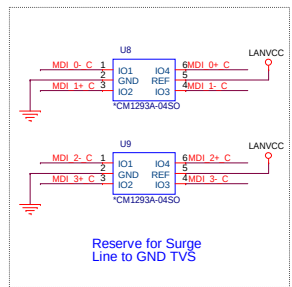
LAN



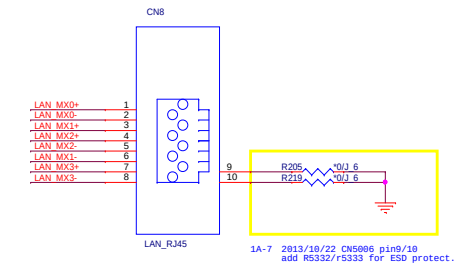
Power trace tracking



Transformer



RJ45 Connector

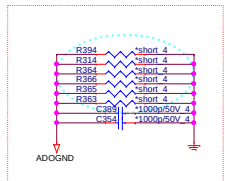


1B-6 2013/12/18 Change CN12 USB2.0 port to port0.



Size	Document Number	Rev
	USB Port/ DB	1A
Date:	Tuesday, April 08, 2014	Sheet 27 of 47

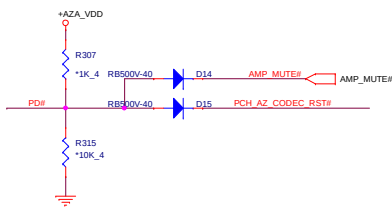
Codec(ADO)



Codec PWR 5V(ADO)



Mute(ADO)



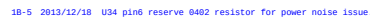
Internal Speaker



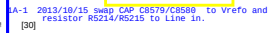
Grounding circuit(ADO)



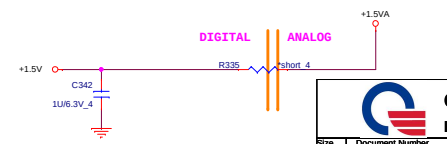
D-Mic



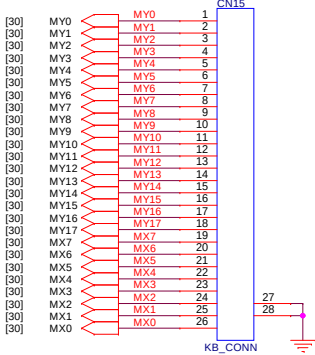
Universal Audio Jack



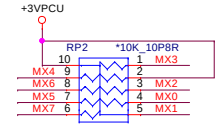
Codec PWR 3V/1.5V(ADO)



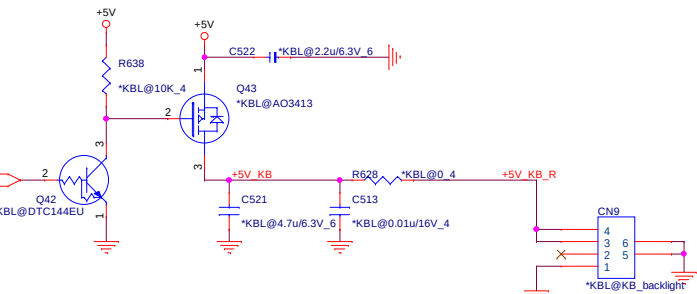
K/B (KBC)



1A-7 2013/10/22 change CN24 pin define based on spec.
1A-8 2013/10/22 change CN24 pin define based on spec based on ZRQ.



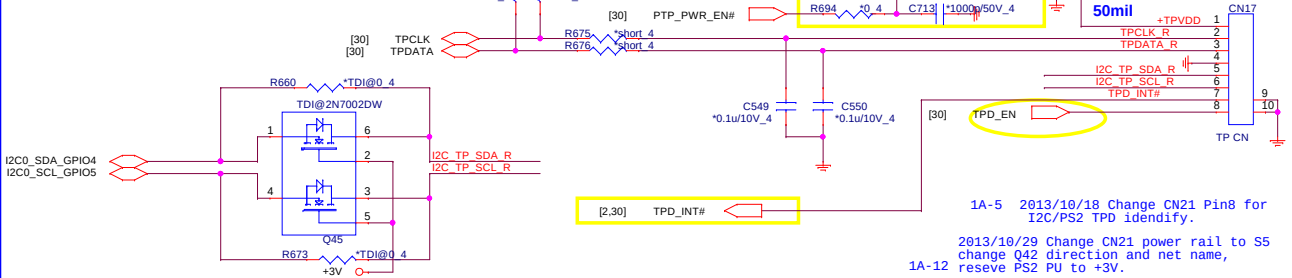
KB_BL LED (KBC)



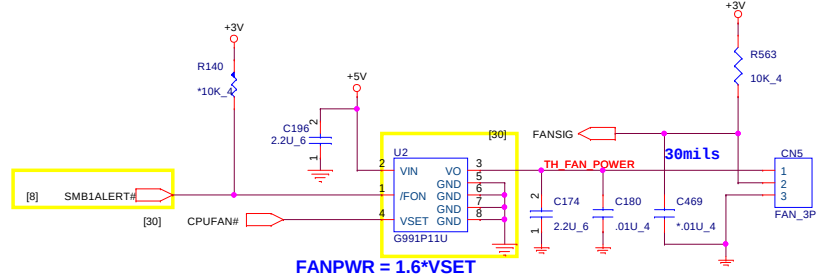
1A-7 2013/10/22 change CN25 pin define for spec.
1A-8 2013/10/23 change CN25 footprint.

TOUCHPAD BOARD CONN (TPD I2C/PS2 co-lay)

TPD->100kHz, TS=400KHz
Intel design guide suggestion
MCP PIN 10u.
Per inch 3u TS=3x5inch
400kHz10-100u =2.4-0.4k.
100KHz 10-100u=9k-1k.

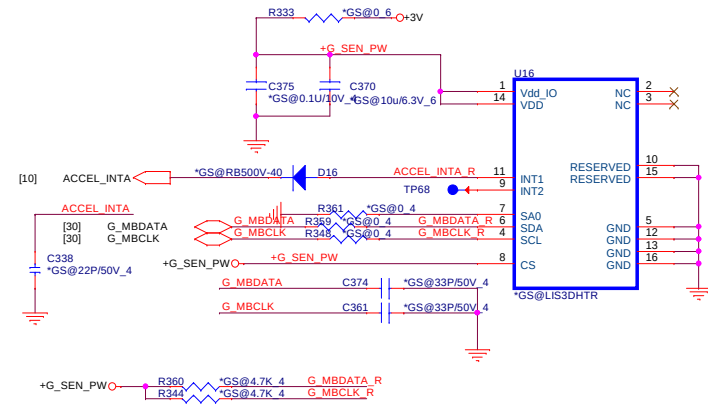


CPU FAN (THM)

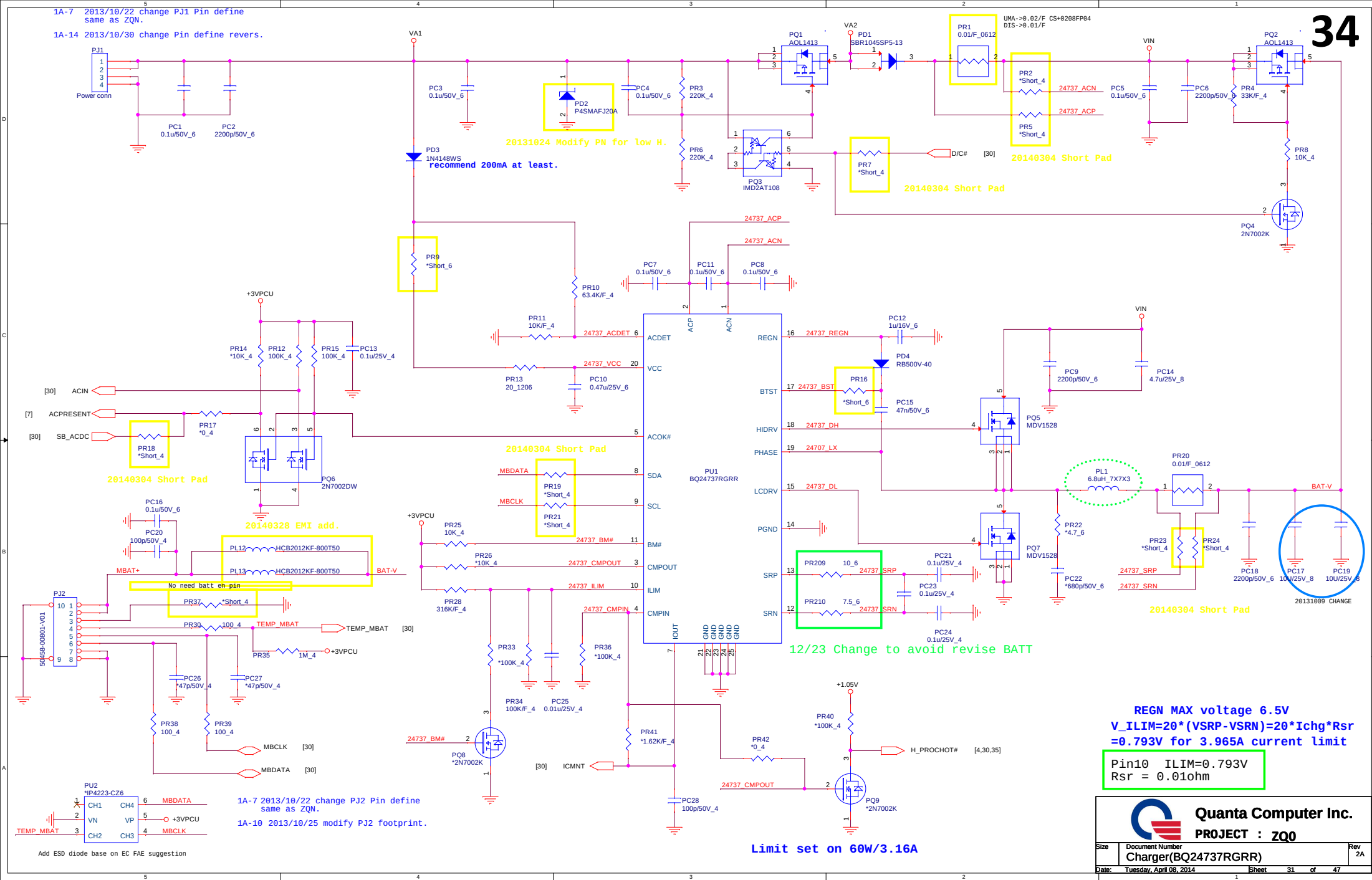


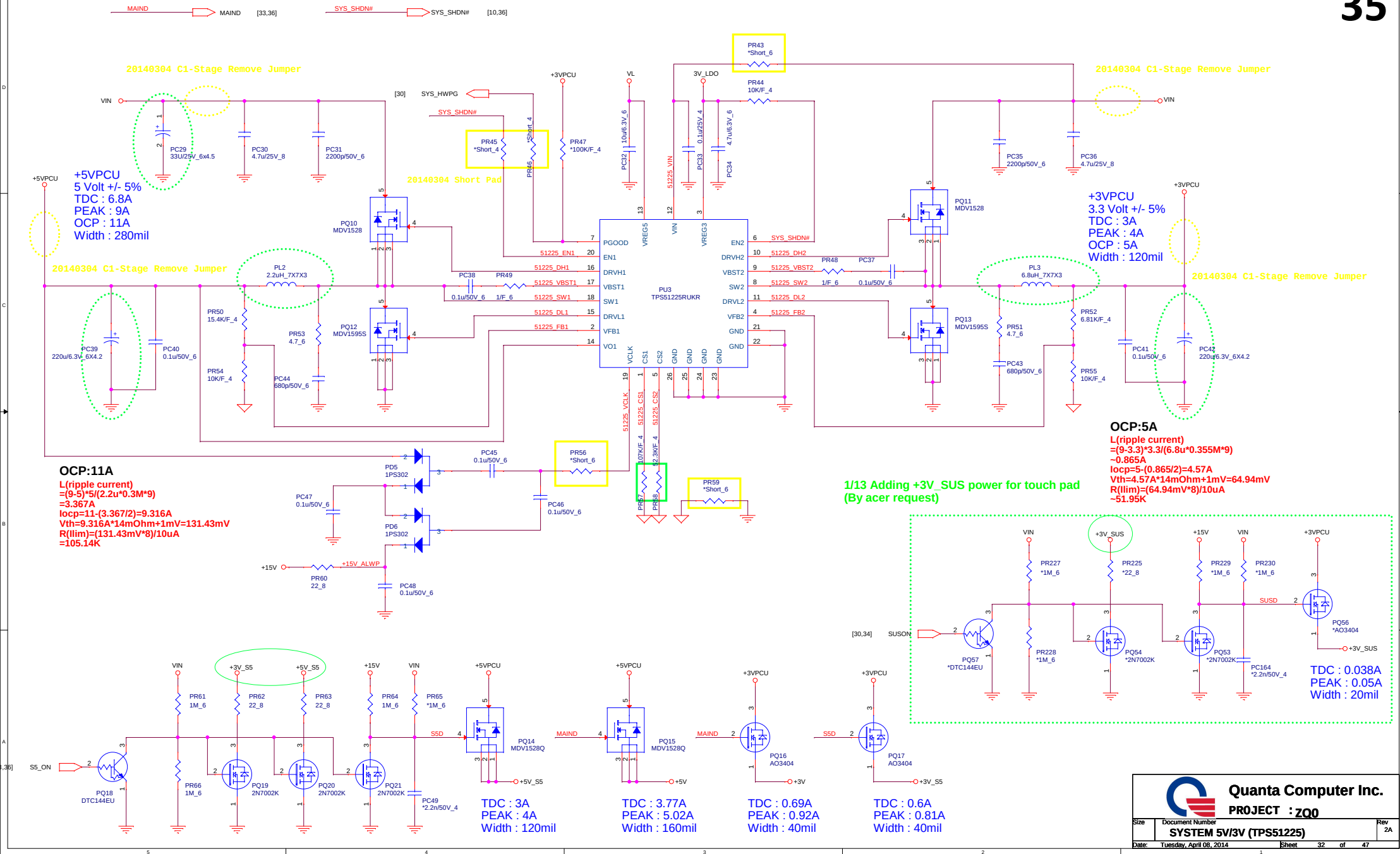
1A-12013/10/15 change pin define and add pwm IC U17.
1A-42013/10/17 Change U17 to G991P11U and PU U17 pin1.
1A-92013/10/24 Add alert on U17.1 for CPU thermal tempature.
1A-13 2013/10/31CN15 Pin2/3 swap.

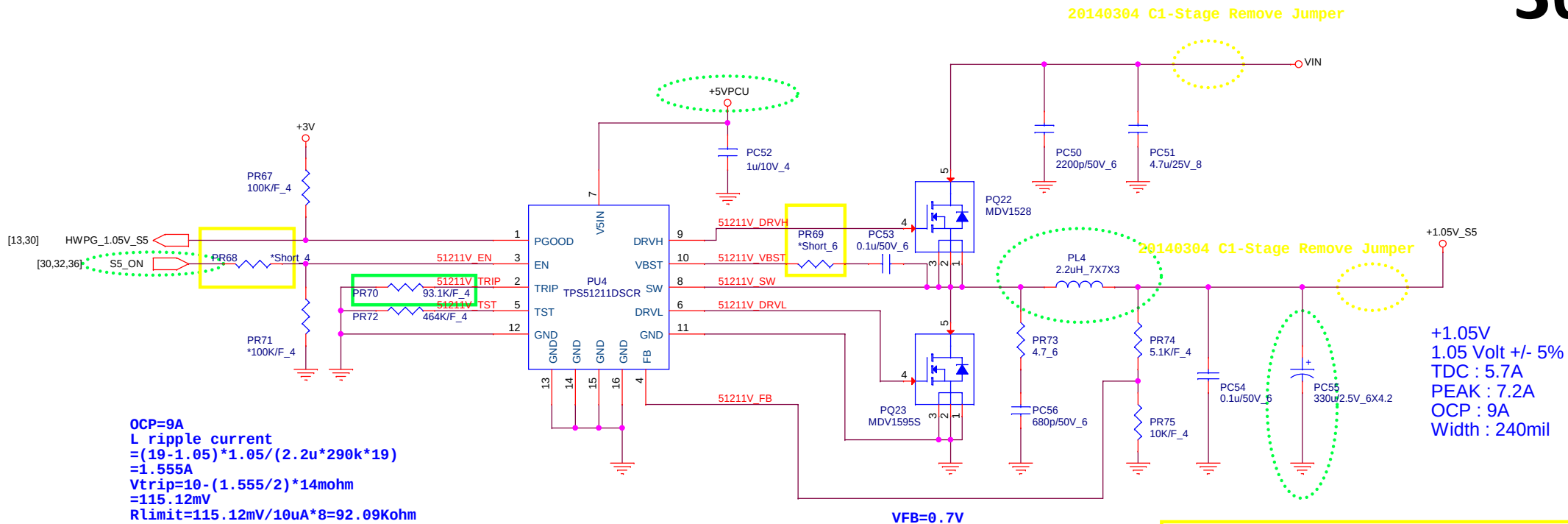
Accelerometer Sensor(reserve only)



1A-7 2013/10/22 change PJ1 Pin define same as ZQN.
1A-14 2013/10/30 change Pin define revers.







OCP=9A
L ripple current

$$= (19 - 1.05) \times 1.05 / (2.2 \times 290 \times 19)$$

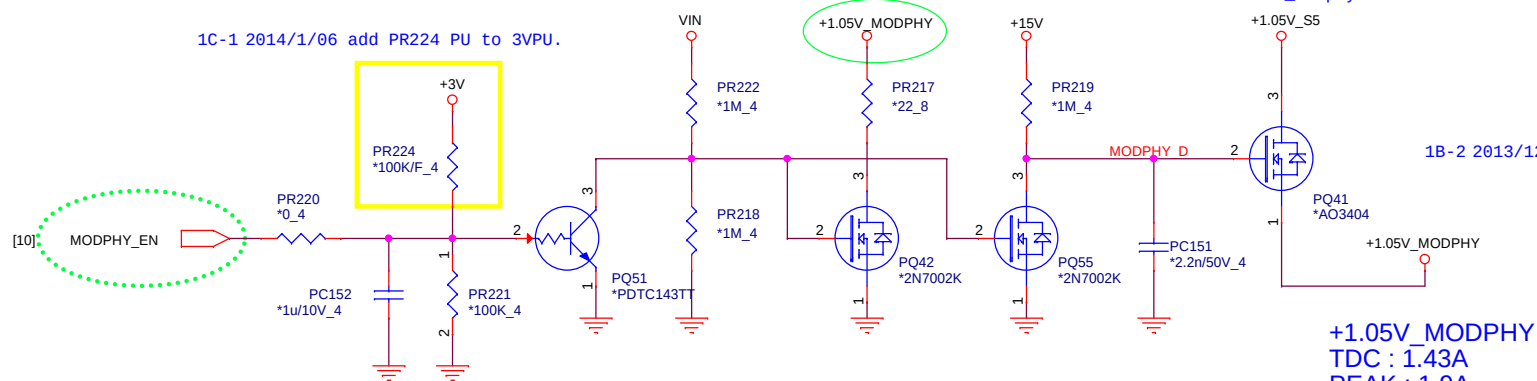
$$= 1.555A$$

$$V_{trip} = 10 - (1.555/2) \times 14m\Omega$$

$$= 115.12mV$$

$$R_{limit} = 115.12mV / 10\mu A \times 8 = 92.09K\Omega$$

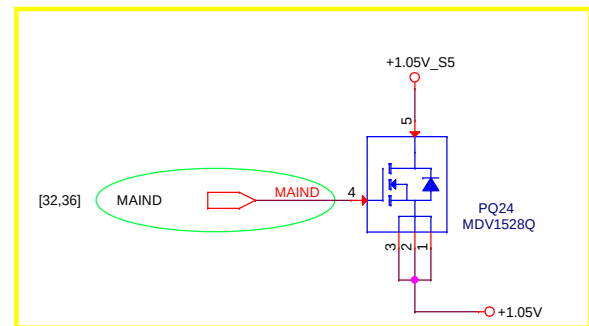
1C-1 2014/1/06 add PR224 PU to 3VPU.



1C-1 2014/1/06 add 0ohm pass 1.05V_Modphy to 1.05V.

1B-2 2013/12/03 change PQ24 to DFN 3x3 size

+1.05V_MODPHY
TDC : 1.43A
PEAK : 1.9A
Width : 80mil



TDC : 2.4A
PEAK : 3.2A
Width : 100mil

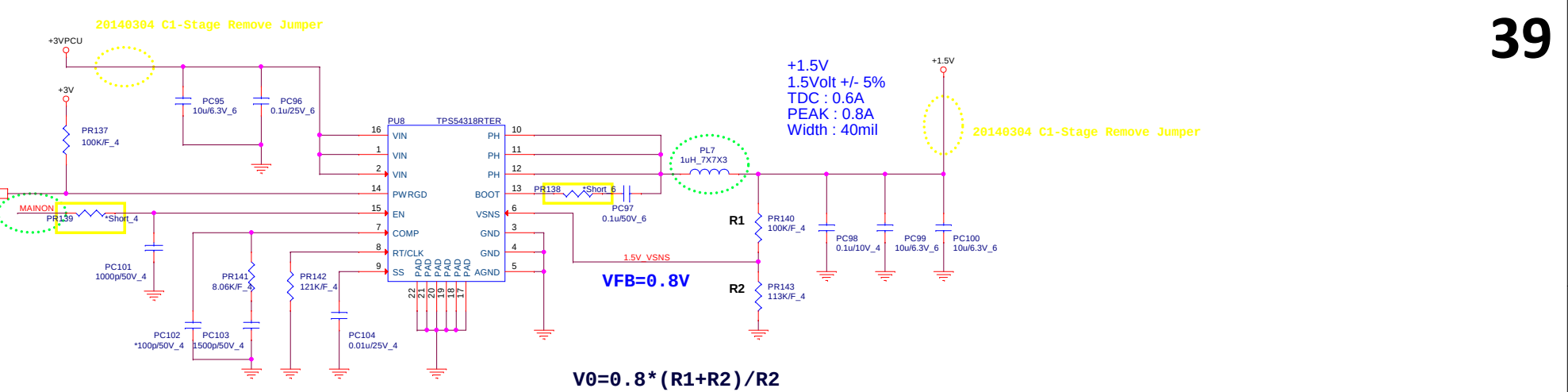


Quanta Computer Inc.
PROJECT : ZQ0

Size	Document Number	Rev
	+1.05V_S5 (TPS51211)	2A

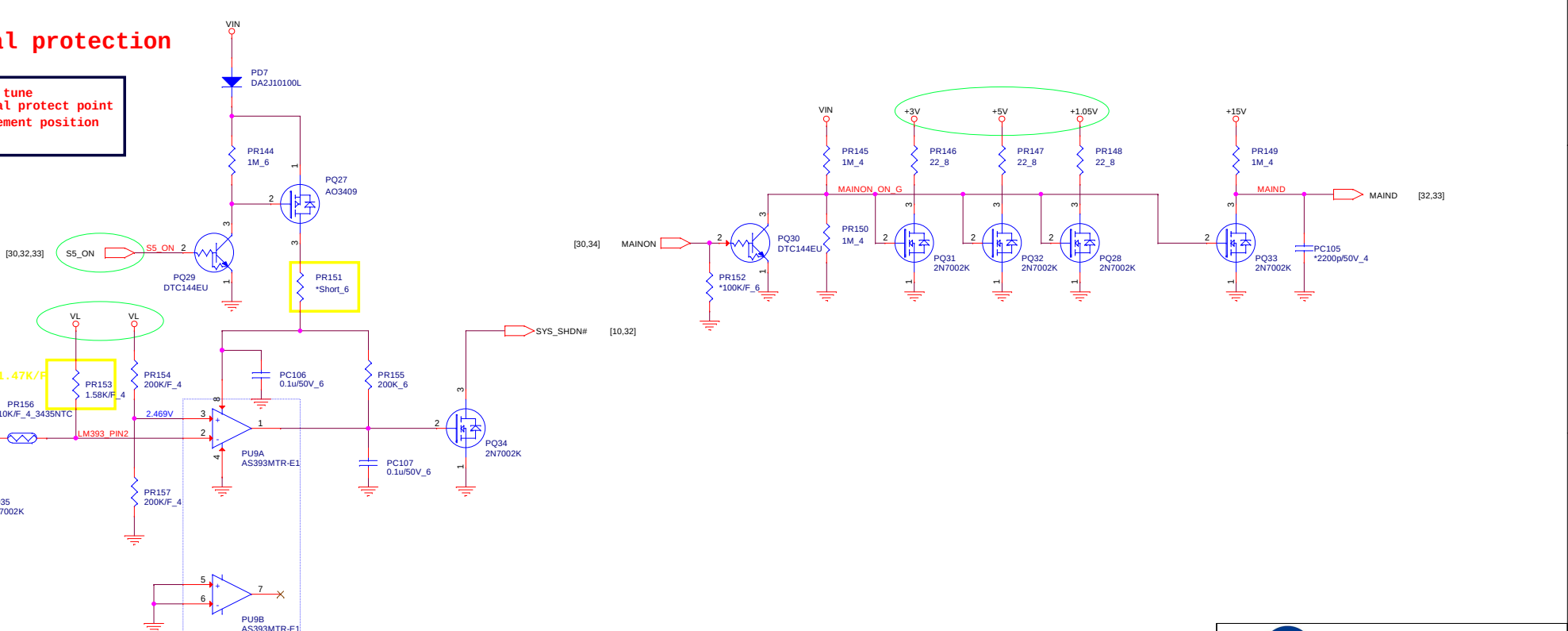
Date: Tuesday, April 08, 2014 Sheet 33 of 47





l protection VIN

Diagram illustrating the connection of the DA2J10100L component (PD7) to the PR144 component. The DA2J10100L is connected to the PR144 component via a connection point labeled "tune al protect point element position".



For EC control thermal protection (output 3.3V)



Standby Function

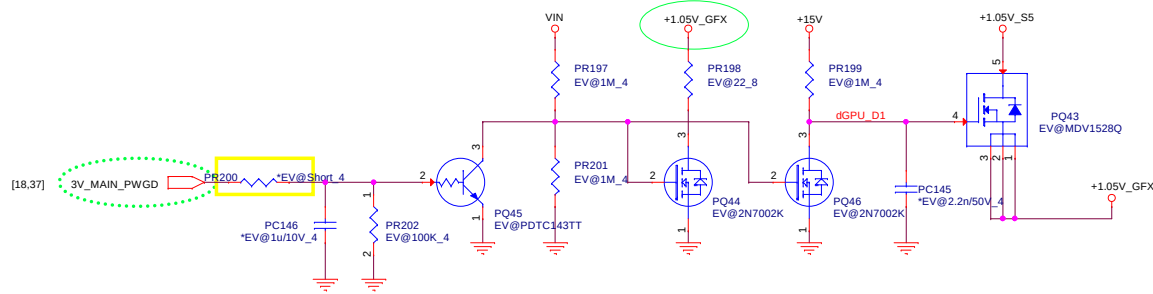
20131018 no need standby function

Parallel

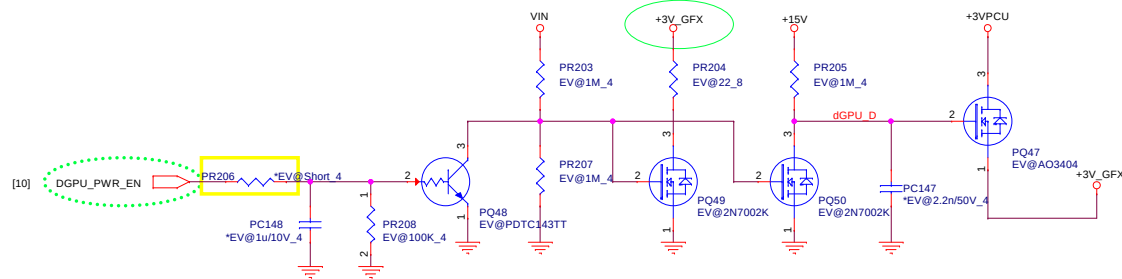


+VGPU_CORE
Countinue current:26A
Peak current:60A
OCP:75A
FSW:300KHz
L/L=0mV/A

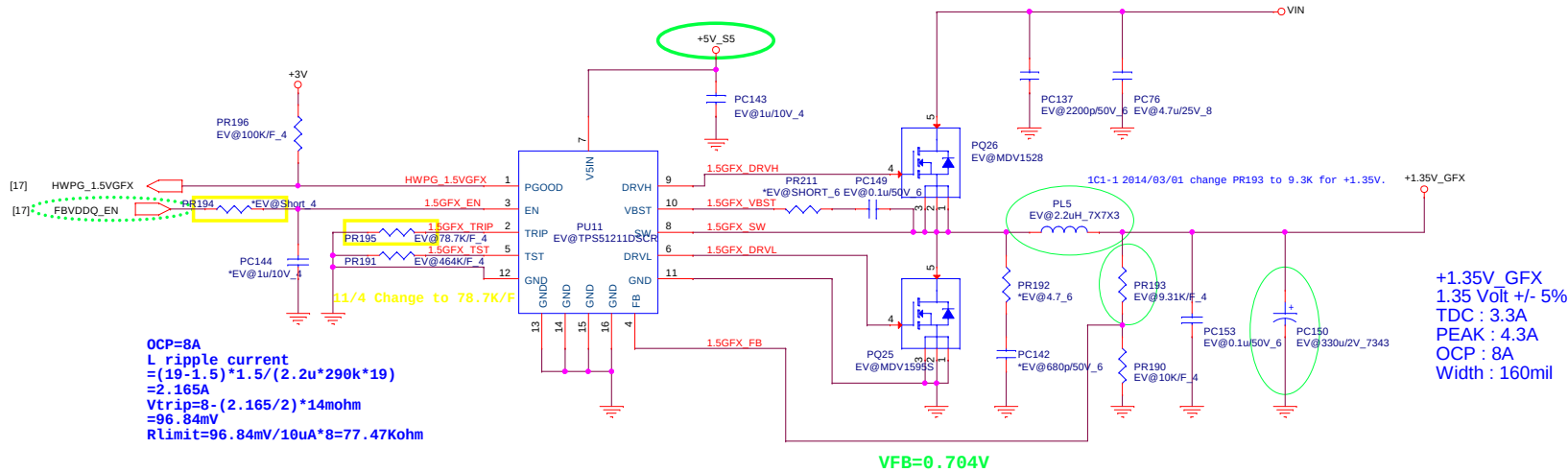
[16,17,18] +1.05V_GFX
[17,20,27] +1.35V_GFX
[16,17,18,19,30] +3V_GFX



+1.05V_GFX
TDC : 1.73A
PEAK : 2.3A
Width : 80mil



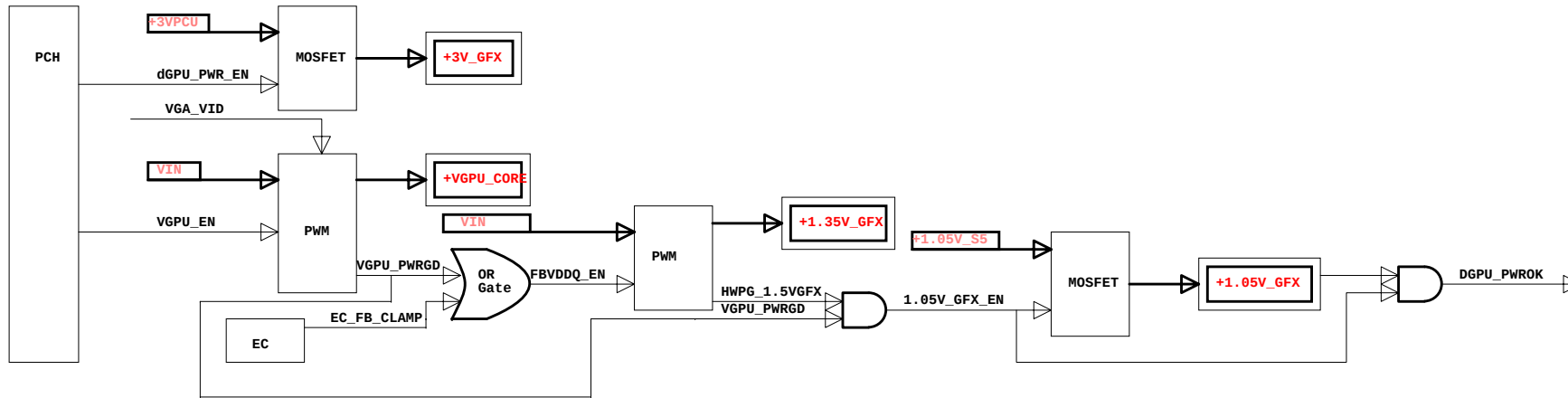
+3V_GFX
TDC : 0.17A
PEAK : 0.23A
Width : 20mil



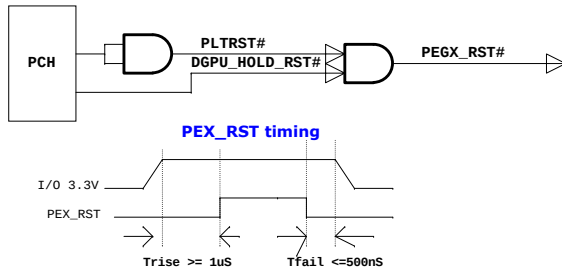
Quanta Computer Inc.
PROJECT : ZQ0

Size	Document Number	Rev
	+1.35V_GFX/+1.05V_GFX/+3V_GFX	2A
Date:	Tuesday, April 08, 2014	Sheet 38 of 47

VGA power up sequence

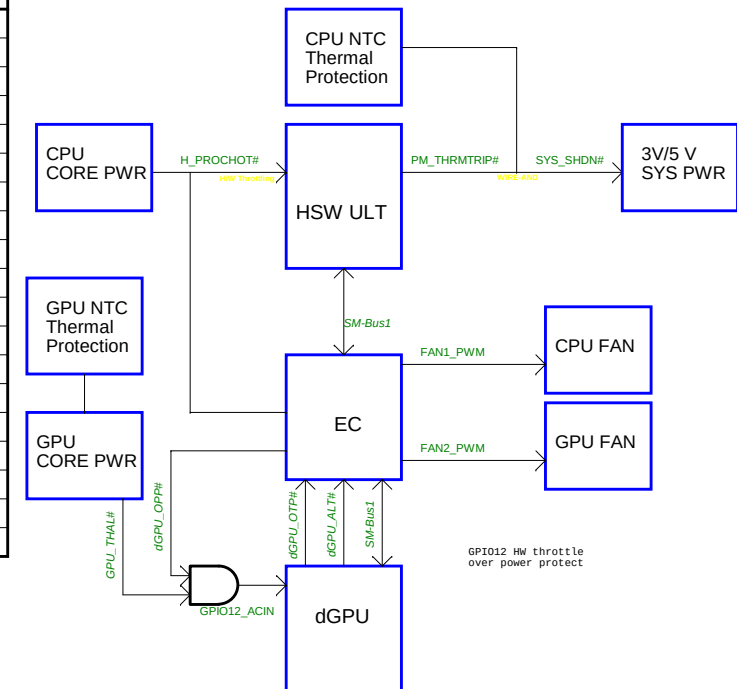


VGA Reset



POWER PLANE	VOLTAGE	DESCRIPTION	CONTROL SIGNAL	ACTIVE I
VIN	+10V~+19V	MAIN POWER	ALWAYS	ALWAYS
+3V_RTC	+3V~+3.3V	RTC POWER	ALWAYS	ALWAYS
+3VPCU	+3.3V	EC POWER	ALWAYS	ALWAYS
+5VPCU	+5V	USB CHARGE POWER	ALWAYS	ALWAYS
+15V	+15V	CHARGE PUMP POWER	ALWAYS	ALWAYS
+3V_S5	+3.3V	LAN/BT POWER	S5_ON	S0-S5
+5V_S5	+5V	USB POWER	S5_ON	S0-S5
+5V	+5V	HDD/SPK/HDMI POWER	MAINON	S0
+3V	+3.3V	PCH/GPU/Peripheral component POWER	MAINON	S0
+1.35VSUS	+1.35V	CPU/SODIMM/MD POWER	SUSON	S0-S3
+DDR_VTT_RUN	+0.675V	SODIMM/MD Termination POWER	MAINON	S0
LCDVCC	+3.3V	LCD POWER	LVDS_VDDEN	S0
+1.5V	+1.5V	MINI CARD/NEW CARD POWER	MAINON	S0
+1.05V	+1.05V	PCH CORE VCCST POWER	MAINON	S0
+VCCIN	variation	CPU CORE POWER	VRON	S0
+VGPU_CORE	variation	External GPU POWER	VGPU_EN	S0
+3V_GFX	+3.3V	External GPU POWER	dGPU_PWR_EN	S0
+1.35V_GFX	+1.35V	External GPU POWER	FBVDDQ_EN	S0
+1.05V_GFX	+1.05V	External GPU POWER	1.05V_GFX_EN	S0

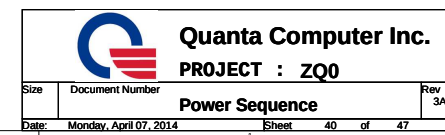
Thermal Follow Chart

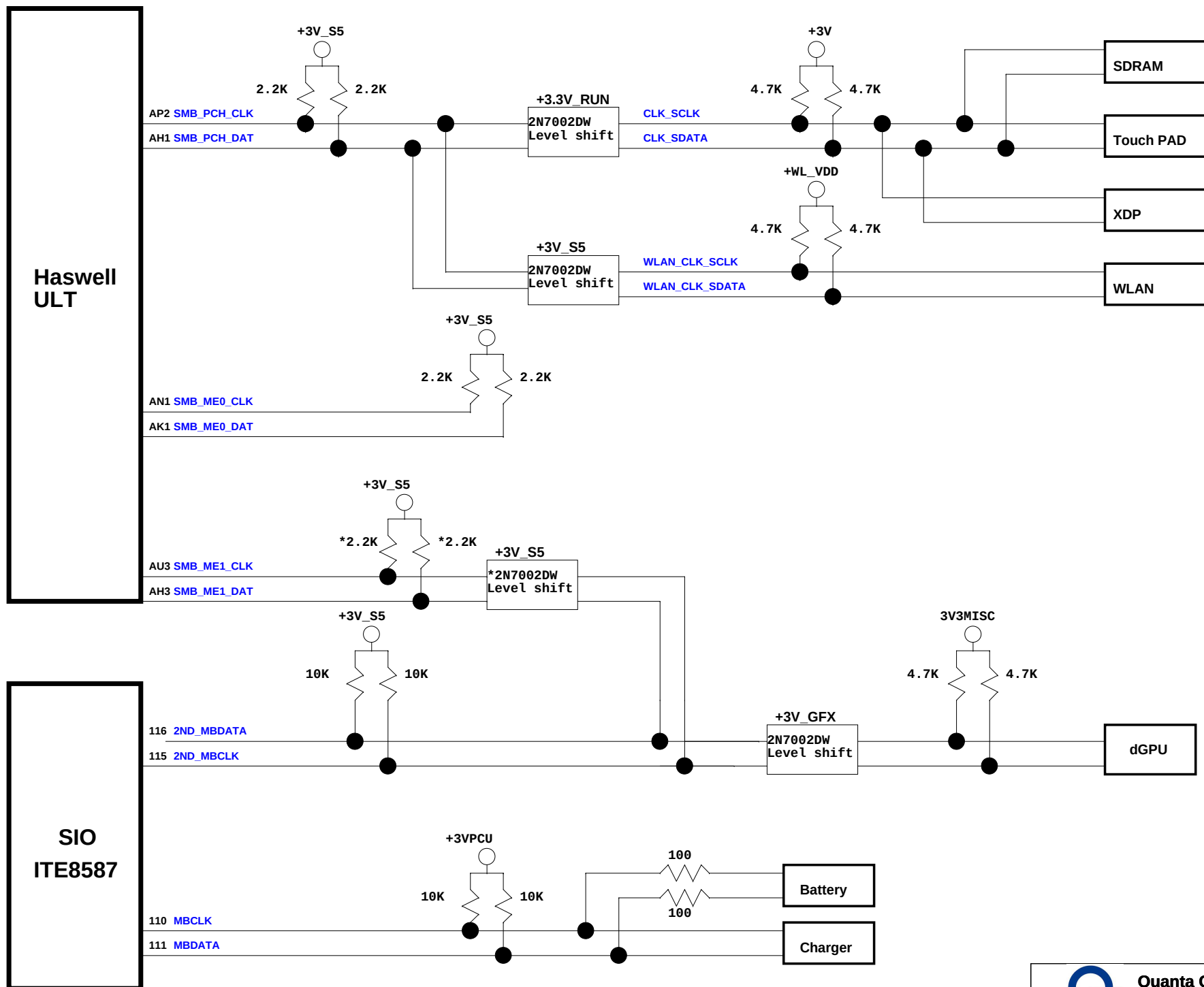


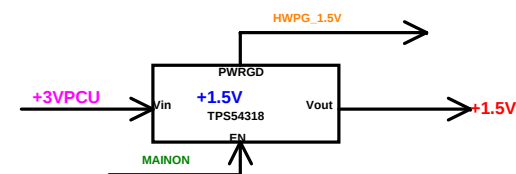
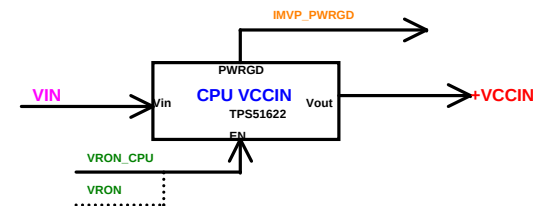
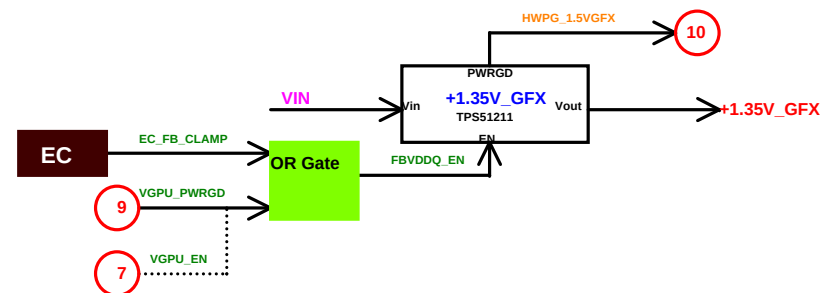
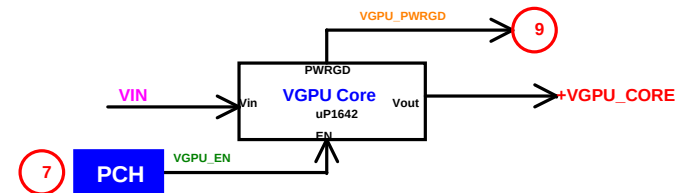
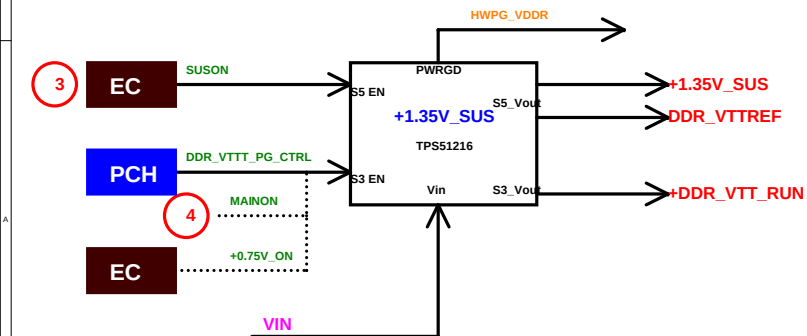
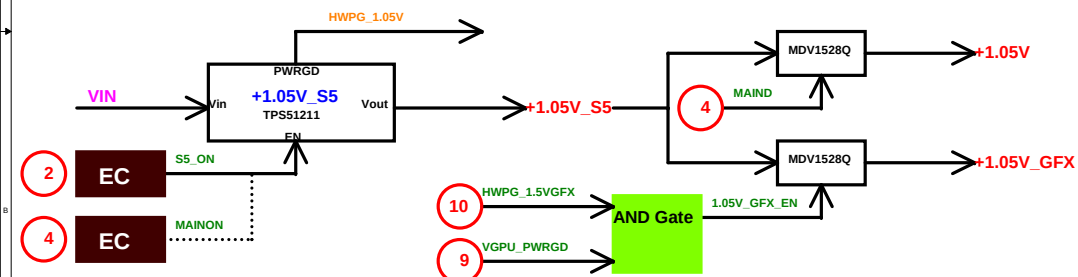
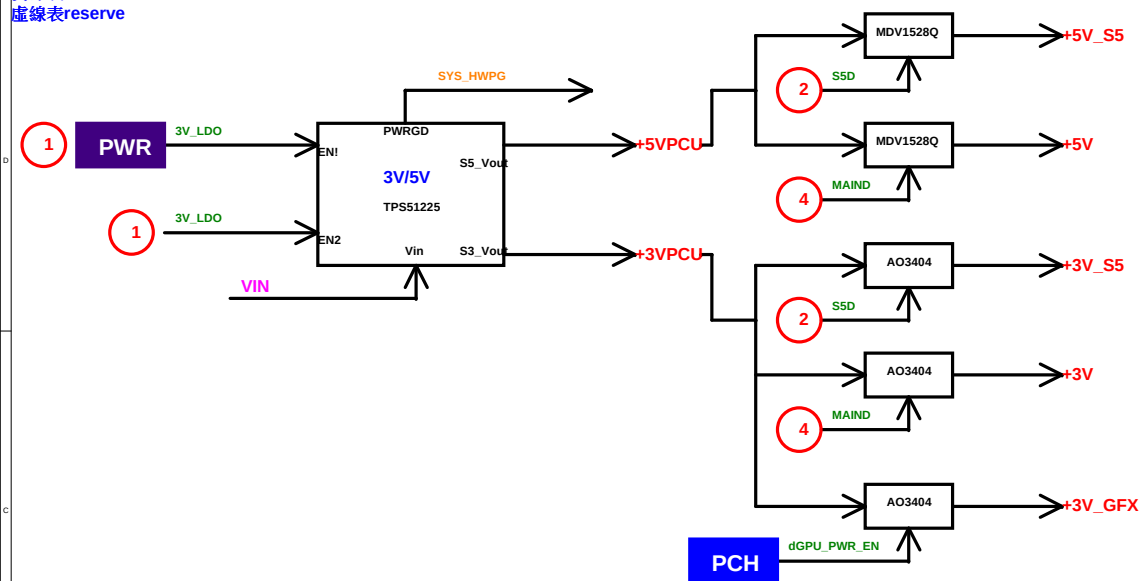
```
dGPU_OPP# EC notify HW throttle over power protect
dGPU_ALT# for ADPS circuit to inform EC NV dGPU VPS Alert
dGPU_OTP# VGA thrmtrip# => inform EC over temperature protect
```

Support Deep Sx

43





實線表default
虛線表reserve

Model	Version	CHANGE LIST				
ZQ0	1A-1	1	2013/10/15 change pin define and add pwn IC.(page31)			
		2	2013/10/15 Change VGA ITE solution to NXP.(page 23)			
		3	2013/10/15 power board CN change to 6pin.(Page 23)			
		4	2013/10/15 U5017.12 change 27M crystal to VGA IC.(Page 23)			
		5	2013/10/15 U5017.14 add power rail +3V_RTC.(page23)			
		6	2013/10/15 strap0 R672 DG 50k. PU.(Page 19)			
		7	2013/10/15 Change AND gat to Q63 D-MOS.(Page 19)			
		8	2013/10/15 change pin define and add pwn IC U17.(page 46)			
		9	2013/10/15 for GC6 stuff R220R103R220R102.am stuff Q24/Q26/R227R1011. (Page19)			
		10	20131015 For GC6 NV DG GC6_FB_EN PD.(Page10)			
		11	2013/10/15 following up acer define and swap USB3 and USB2 port.(Page9)			
		12	2013/10/15 swap CAP C8579/C8580 to Verbs and resistor R5214/R5215 to Line in.(Page30)			
		13	2013/10/15 U27.30/U27.31 del fan Pwm signal.(Page32)			
		14	20131015 change LVDS/USB3R45/FAN/TPD/USB DB CN/DC-IN CN/Power Button/Cardreader/KB BLK CN/Power board, footprint.			
	1A-2	1	2013/10/16 JDM01 Swap M_R_DQ52/M_R_DQ53 and swap M_R_DQ502/M_R_DQ503.(page13)			
		2	2013/10/16 JDM06 Chage net name M_R_DQ5047-01 to M_A_DQ5047-01.(page14)			
		3	2013/10/16 Add RTC charge circuit.(page9)			
		4	2013/10/16 RTL1 Chage +3V_RTC_0 to VCCTC_2.(page6)			
		5	2013/10/15 change power rail from +3V_RTC_0 to VCCTC_2.(page23)			
	1A-3	1	2013/10/16 change R5285 from 330 to 100ohm for charge RTC battery.(page23)			
		2	2013/10/16 2013/10/16 U50 add 0ohm R5322/R5323 for SMBus reserve for FW burning.(page 23)			
		3	2013/10/16 U50 pin24/25 add 33ohm for HSYNCSVSYNC.(page23)			
		4	2013/10/16 U50.37 add 10 ohm for test pin avide i2c impact.(page 23)			
		5	2013/10/16 U24 ball R4/G2 BIOS suggestion change SMU3CT to GPIO0-15.(page 10)			
		6	2013/10/16 Add U34 flash 4M ROM reserve for ZQ0D.(page6)			
		7	2013/10/16 change SMBus VGA to PCH SMLBCLK/SMLB0DATA.(page 8)			
	1A-4	1	2013/10/17 Change EC pin define for 2014 GPIO table.(page32)			
		2	2013/10/17 Change A0 short pad to resistor.(A0)			
		3	2013/10/17 Change U17 to G991P11U and PU U17 pin1.(page31)			
		4	2013/10/17 Remove Q25/R231/R232 because not support GPIO9 for ADPS circuit to inform EC NV dGPU VPS Alert.(page19)			
		5	2013/10/17 remove Q5020 no IOAC support.(page26)			
		6	2013/10/17 remove R5224/R5225/R5226 no IOAC support.(page26)			
		7	2013/10/17 remove WLAN_OFF no IOAC support.(page26)			
		8	2013/10/17 Del U22 because no support IOAC.(page32)			
	1A-5	1	2013/10/18 Change CN21 Pin0 for I2C/PS2 TPD identify.(Page31)			
		2	2013/10/18 Change VGA NXP solution to ITE.(page23)			
		3	2013/10/18 design change R5293 from 22ohm to 33ohm.(page23)			
		4	2013/10/18 Change Touch screen power rail from 5V to 3V.(page24)			
		5	2013/10/18 add 0ohm short TP interrpt pin.(page24)			
		6	2013/10/18 change U27.B7 for Touch pad ID for I2C/PS2 solution switch.(page 32)			
	1A-6	1	2013/10/18 Change Q63 to MOS.(page19)			
		2	2013/10/21 reversal PEG las for layout.(page9)			
		3	2013/10/21 Del AP/WORK.(page5)			
		4	2013/10/19 Swap DDR to-ddim pin for layout request.(page14,15)			
	1A-7	1	2013/10/22 change CN24 pin define based on spec.(page31)			
		2	2013/10/22 change CN25 pin define for spec.(page31)			
		3	2013/10/22 Change CN4 to 6pin.(page23)			
		4	2013/10/22 change Y2004 to +1.10PPM.(page23)			
		5	2013/10/21 add R5331 between 3V_GFX and 3V_MIN for not GC6 support.(page20)			
		6	2013/10/22 change P11 Pin define same as ZQN.(page33)			
		7	2013/10/22 Change L.GND to GND.(page28)			
		8	2013/10/22 CNS006 pin9/10 add R5332/r5333 for ESD protect.(page20)			
		9	2013/10/22 Change CNS009/CNS013/Y7 footprint.			
	1A-8	1	2013/10/22 change CN24 pin define based on spec based on ZRQ.(page31)			
		2	2013/10/23 change CN25 footprint.(page31)			
		3	2013/10/23 Change DIMM1_SAB/SA1 to DIMM0_SAB/SA1.(page14)			
		4	20131022 Change GPIOB3/04 GPU GC6 pin to GPIO2/3.(page10)			
		5	2013/10/21 Swap PEG to manual mode.(page9)			
		6	2013/10/23 remove R5017 5V is duplicate.(page27)			
		7	2013/10/23 remove C3056 is duplicate.(page27)			
		8	2013/10/23 change CN000/CN25 footprint to match DXF.			
		9	2013/10/23 add screw hole footprint.			
	1A-9	1	2013/10/17 Change power LED from +3VPCU to +3V_S5.(page27)			
		2	2013/10/24 add 100kohm on U5011 pin 26 with C5164 for discharge.(page30)			
		3	2013/10/23 add 10k for vendor suggestion.(page28)			
		4	2013/10/23 add R5335 isolate SLR0600 NC.(page23)			
		5	2013/10/24 Add alert on U17.1 for CPU thermal tmprture.(page31).			
	1A-10	1	2013/10/25 remove 1.35GFX_PDG.NET.(page20)			
		2	2013/10/25 remove 1.35GFX_PDG.(page20)			
		3	2013/10/25 remove 1.35GFX_POWER.(page20)			
		4	2013/10/25 reserve AV2 ball to GND.(page6)			
		5	20131025 reserve A5 ball to 100k PU 3VPCU.(page6)			
		6	2013/10/25 modify P12 footprint.(page33)			
		7	2013/10/25 change LED from 3pin to 4pin for acer request.(page27)			
	1A-11	1	2013/10/28 Change U5013.7 from +3V_S5 to +3V.(page23)			
		2	2013/10/28 change LED from 4pin to 3pin and power LED to +3VPCU.(page27)			
		3	2013/10/28 U5013 Pin0,1,5,20 left NC.(page23)			
	1A-12	1	2013/10/29 Change CN21 power rail to 5V.change Q42 direction and net name reserve PS2 PU to +3V.(page31)			
		2	20131029 Change GPIO45 to PU 5S,ahble GPIO5B one is GPIO56.(page10).			
		3	2013/10/28 reserve AV2 ball to TP.(page6)			
		4	20131028 reserve A5 ball to TP.(page6)			
		5	2013/10/29 Change CNS000 to S8 of S8mus.(page26)			
		6	2013/10/29 Swap U27 pin2 and pin120,add U27.08 for touch pad interrpt.(page32)			
	1A-13	1	20131030 add touch pad interrpt pin on gpio13.(page10)			
		2	2013/10/30 move Q42 to page02 change U24,U7 net name.(page2).			
		3	2013/10/30 CNS002.6 add USB touch screen on/off pin to EC.(page24)			
		4	2013/10/29 add U27.35 for touch pad on/off.(page32)			
DOC NO.	PROJECT MODEL :	ZQ0	APPROVED BY:		DATE:	
	PART NUMBER:		DRAWING BY:		REVISION:	
			 Quanta Computer Inc. PROJECT : ZQ0 Change list4			

Model	Version	CHANGE LIST															
ZQ0	1B-2	1	2013/12/04 change PQ24 to DEN 3x3 size.(page35)														
		2	2013/12/04 Change Ca14 PN and footprint.(page30).														
		3	2013/12/04 change LED from 1pin to 4pin.(page27)														
		4	2013/12/4 change cN6 to 4pin.(page23)														
		5	2013/12/04 change GPIO36/GPI037 to PU.(page9)														
	1B-3	1	2013/12/10 change Ca20 Pin define.(page25)														
		2	2013/12/10 change Q3.3 from +3V to +3VPCU.(page22).														
		3	2013/12/10 change CN6 footprint.(page23)														
	1B-4	1	2013/12/12 Remove U9 Green CLK circuit.(page21)														
	1B-5	1	2013/12/17 Change CN14 pin define.(page20)														
		2	2013/12/17 Change R8051 to 0402 size.(page16)														
	1B-6	1	2013/12/18 Change USB port USB3.0 to port0,USB2.0 to port1 and port3,3Fingerprint to usb port2.														
		2	2013/12/17 Change R8051 to 0402 size.(page16)														
		3	2013/12/18 U34 pin6 reserve 0402 resistor for power noise issue.(page28)														
	1B-7	1	2013/12/20 add U29 VSYNC and HSYNC by pass resistor.(page22)														
		2	2013/12/20 Change +3VPCU to +3V_S5 nom deep ss.(page10).														
		3	2013/12/20 del c0521 and R8391.(page16)														
	1C-1	1	2014/1/06 add 0ohm pass 1.05V_Modify to 1.05V.(page33)														
		2	2014/1/06 add PR224 PU to 3V.(page33).														
		3	2014/1/06 Change R351/R388 from 470hm to 650hm base on FAE request.(page28)														
		4	2014/01/10 Remove U29 and add U40 and U41.(page25)														
	1C-2	1	2014/01/13 Change TP power rail from +3V_S5 to +3V_SUS.(page29)														
		2	20140113 PCU/PU12 change footprint for SMT request.(page35).														
		3	2014/01/13 change CN14 data net name and add C678~C681.(page25)														
		4	2014/01/13 add R670/R677 PU and R679 PD for ICT.(page19)														
		5	2014/01/13/13 Adding +3V_SUS power for touch pad (acer request).(page32)														
	1C-3	1	2014/01/14 change R654 to 0ohm.(page27)														
		2	2014/01/14 Change Ca11 Footprint.(page24).														
	1C-4	1	2014/01/15 reserve TP power rail +3V_S5.(page29)														
		2	2014/01/15 TTPM CO-Lay navotant(page21).														
		3	2014/01/15 SWAP PCIE LAN TX single.(page26).														
	1C1-1	1	2014/03/01 change PR193 to 9.3K for +1.35V.(page38)														
		2	2014/02/17 Add U11,50 GPIOs for PTP power on function.(page30).														
		3	2014/02/17 Add Q47 for PTP power EN and soft start R694/C713 and C712/C686.(page29)														
		4	2014/02/06 change Blue LED power rail to +5VPCU and add ESD and Change LED to lite-on and R379~R20,R375~680 base on test result.(page25)														
		5	2014/02/6 add VCA_ALERT# PU 10K for FAE request.(page19)														
		6	2014/02/19 add R602 for SUSPWRACK# to EC.(page07)														
		7	2014/03/01 Change 0ohm to short pad.														
		8	2014/03/01 link L29 to +3V directly(meet 1VDDO vs OVDD sequence)(page 21)														
	1C1-2	1	2014/03/00 add R696/R697 PU.(page30)														
		2	2014/03/00 Change PU12 footprint to mt23 and add VC2/VC1 change C307 to 3528.(page27).														
		3	2014/03/00 Remove PCIE wake and stuff R642, no stuff Q44.(page24)														
		4	2014/03/11 Add R690 for TS_EN short TP_INT,for issue debug.(page22)														
DOC NO.	PROJECT MODEL	ZRQ	APPROVED BY:	DATE:													
	PART NUMBER:		DRAWING BY:	REVISION:													
 Quanta Computer Inc. PROJECT : ZQ0 Change History <table><tr><td>Rev</td><td>Rev</td><td>Rev</td><td>Rev</td><td>Rev</td><td>Rev</td></tr><tr><td>01</td><td>02</td><td>03</td><td>04</td><td>05</td><td>06</td></tr></table>						Rev	Rev	Rev	Rev	Rev	Rev	01	02	03	04	05	06
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